NOAA Technical Memorandum NMFS-XXX-##

Analyses to support the Dolphinfish Management Strategy Evaluation in the U.S. Atlantic

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January 2023

U.S. DEPARTMENT OF COMMERCE

National Oceanic and Atmospheric Administration
National Marine Fisheries Service
Northwest Fisheries Science Center

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1 Introduction

Here we will be displaying various data summaries and analyses related to dolphin management in the U.S. South Atlantic.

1.1 Details

The documentation of this data exploration was developed in R, using R Studio and Quarto.

2 NOAA Fisheries recreational landings data

Here we will partition recreational landings data from NOAA Fisheries' Marine Recreational Information Program (MRIP), a standardized, peer-reviewed suite of recreational fishing surveys used to produce catch and effort estimates for quota monitoring and stock assessment in the United States. MRIP is a standardized survey suite that employs a complex statistical sampling design stratified across time, regions of the coast, and different fishing modes, which allows the intercept data to be scaled up by effort estimates to calculate total landings.

2.1 Data access and upload

Data are downloaded directly from the Miami server using the files provided to SERO for ACL monitoring. The current location is: M://SFD/SECM-SFD/ACL/2025_Aug19_-MRIP_FES/. The landings are in FES MRIP and the units are in pounds whole weight.

```
# clear workspace
rm(list = ls())

# read in data file
load("data/mrip_fes_rec81_25wv2_23June25.RData")  # most recent file on SFD server
head(dat)
```

```
  REC_ACL_MRIP_FES_ID YEAR MONTH WAVE SUB_REG NEW_ST NEW_STA NEW_MODE NEW_MODEN
1            73407425 1989    NA    4       6      9      NC        4      Priv
2            73407426 1989    NA    4       6      9      NC        4      Priv
3            73407427 1989    NA    4       6      9      NC        4      Priv
4            73407428 1989    NA    4       6      9      NC        4      Priv
5            73407429 1989    NA    4       6      9      NC        4      Priv
6            73407430 1989    NA    4       6      9      NC        4      Priv
  AREA_X NEW_AREA  NEW_AREAN FL_REG NC_REG   DS               NEW_SCI
1      5        5    Inshore     NA      S MRIP Centropristis striata
2      5        5    Inshore     NA      N MRIP Centropristis striata
```

```
3      2        2  Ocean>3mi     NA      N MRIP Centropristis striata
4      2        2  Ocean>3mi     NA      S MRIP Centropristis striata
5      1        1 Ocean<=3mi     NA      N MRIP Centropristis striata
6      1        1 Ocean<=3mi     NA      S MRIP Centropristis striata
         NEW_COM    SP_CODE SPECIES_ITIS SPECIES        AB1          B2
1 black sea bass 8835020301       167687         34181.4033 121019.1368
2 black sea bass 8835020301       167687           795.8612   4445.0527
3 black sea bass 8835020301       167687           549.5390   2919.4840
4 black sea bass 8835020301       167687         71823.2390  24548.2535
5 black sea bass 8835020301       167687           415.7642     87.5928
6 black sea bass 8835020301       167687         48728.5632  11859.6428
           A         B1 LBSEST_SECWWT LBSEST_SECSOURCE SAMPLE_SIZE_USED
1 31761.4755  2419.9278    11457.8425          srysmwa               41
2   584.4155   211.4457      266.7782          srysmwa               41
3   549.5390     0.0000      484.4319          srysmwa              119
4 33343.1844 38480.0546    63313.9276          srysmwa              119
5   415.7642     0.0000      294.0623          srysmwa               71
6 40585.0152  8143.5480    34464.8156          srysmwa               71
   AVG_LEN   AVG_WGT SRYSMWA_AVG_LEN SRYSMWA_AVG_WGT SRYSMWA_SAMPLE_SIZE
1 203.4572 0.3352069        203.4572       0.3352069                  41
2 203.4572 0.3352069        203.4572       0.3352069                  41
3 296.2590 0.8815243        296.2590       0.8815243                 119
4 296.2590 0.8815243        296.2590       0.8815243                 119
5 261.9949 0.7072816        261.9949       0.7072816                  71
6 261.9949 0.7072816        261.9949       0.7072816                  71
  SRYSMWA_FLAGGED SRYSMW_AVG_LEN SRYSMW_AVG_WGT SRYSMW_SAMPLE_SIZE
1              NA       269.2563      0.7310037                231
2              NA       269.2563      0.7310037                231
3              NA       269.2563      0.7310037                231
4              NA       269.2563      0.7310037                231
5              NA       269.2563      0.7310037                231
6              NA       269.2563      0.7310037                231
  SRYSMW_FLAGGED SRYSM_AVG_LEN SRYSM_AVG_WGT SRYSM_SAMPLE_SIZE SRYSM_FLAGGED
1             NA      259.7694     0.6948228               556            NA
2             NA      259.7694     0.6948228               556            NA
3             NA      259.7694     0.6948228               556            NA
4             NA      259.7694     0.6948228               556            NA
5             NA      259.7694     0.6948228               556            NA
6             NA      259.7694     0.6948228               556            NA
  SRYS_AVG_LEN SRYS_AVG_WGT SRYS_SAMPLE_SIZE SRYS_FLAGGED SRY_AVG_LEN
1      270.955    0.8099641              982           NA    269.1276
```

```
2      270.955   0.8099641             982          NA    269.1276
3      270.955   0.8099641             982          NA    269.1276
4      270.955   0.8099641             982          NA    269.1276
5      270.955   0.8099641             982          NA    269.1276
6      270.955   0.8099641             982          NA    269.1276
  SRY_AVG_WGT SRY_SAMPLE_SIZE SRY_FLAGGED SR_AVG_LEN SR_AVG_WGT SR_SAMPLE_SIZE
1   0.8085211            1703          NA   303.6871   1.050403          46055
2   0.8085211            1703          NA   303.6871   1.050403          46055
3   0.8085211            1703          NA   303.6871   1.050403          46055
4   0.8085211            1703          NA   303.6871   1.050403          46055
5   0.8085211            1703          NA   303.6871   1.050403          46055
6   0.8085211            1703          NA   303.6871   1.050403          46055
  SR_FLAGGED S_AVG_LEN S_AVG_WGT S_SAMPLE_SIZE S_FLAGGED     VAR_B2   VAR_AB1
1         NA  344.4373  1.364082        305089        NA 4072492794 310099094
2         NA  344.4373  1.364082        305089        NA    2866289    258947
3         NA  344.4373  1.364082        305089        NA          0         0
4         NA  344.4373  1.364082        305089        NA  105731413 534702898
5         NA  344.4373  1.364082        305089        NA          0         0
6         NA  344.4373  1.364082        305089        NA   22265768 286196169
        SA_LABEL GOM_LABEL REC_ACL MIGRA_GRP AGG_MODEN JURISDICTION PS_REG
1 Snapper Grouper                1             Private        State      6
2 Snapper Grouper                1             Private        State      5
3 Snapper Grouper                1             Private      Federal      5
4 Snapper Grouper                1             Private      Federal      6
5 Snapper Grouper                1             Private        State      5
6 Snapper Grouper                1             Private        State      6
  COUNCILREG GFMC_FMP SAFMC_FMU LBSEST_SECGWT AGGR_AREA AREA FIRST_MONTH
1          6                  Y     9710.0360             NA          NA
2          5                  Y      226.0832             NA          NA
3          5                  Y      410.5355             NA          NA
4          6                  Y    53655.8708             NA          NA
5          5                  Y      249.2054             NA          NA
6          6                  Y    29207.4709             NA          NA
  LAST_MONTH ALT_FLAG CHTS_CL CHTS_H CHTS_RL CHTS_WAB1C CHTS_WAB1H MODE_FX
1         NA        0       0      0       0          0          0       7
2         NA        0       0      0       0          0          0       7
3         NA        0       0      0       0          0          0       7
4         NA        0       0      0       0          0          0       7
5         NA        0       0      0       0          0          0       7
6         NA        0       0      0       0          0          0       7
  NEW_SEASN SEASON
```

```
1                NA
2                NA
3                NA
4                NA
5                NA
6                NA
```

```
#apply(dat[2:18], 2, table, useNA = "always")

# confirm nomenclature for dolphin and subset by species --------------------------
table(dat$NEW_COM[grep("dolphin", dat$NEW_COM)])
```

```
dolphin
   9377
```

```
table(dat$NEW_SCI[grep("dolphin", dat$NEW_COM)])
```

```
Coryphaena hippurus
               9377
```

```
d <- dat[which(dat$NEW_COM == "dolphin"), ]
#apply(d[2:18], 2, table, useNA = "always")
d <- d[which(d$YEAR < 2025), ]    # take out 2025 because incomplete year
```

2.2 Separate Gulf and Atlantic recreational landings

Now we filter the data set to separate out the Gulf versus Atlantic landings.

```
# codes for WFL regions: 1 (NW FL Panhandle- Escambia to Dixie co.), 2 (SW FL Peninsula- Le
table(d$FL_REG, useNA = "always")
```

```
   1    2    3    4    5 <NA>
 387  110  568 1022  312 6943
```

```r
# subregional codes: 6 is Atlantic and 7 is Gulf
table(d$FL_REG, d$SUB_REG, useNA = "always")
```

```
          4    5    6    7 <NA>
  1       0    0    0  387    0
  2       0    0    0  110    0
  3       0    0    0  568    0
  4       0    0 1022    0    0
  5       0    0  312    0    0
  <NA>   96  585 3836 2426    0
```

```r
unique(d$NEW_STA)
```

```
 [1] "NC"     "FLE"    "FLW"    "SC"     "GA"     "MS"     "AL"     "LA"    
 [9] "TX"     "FLE/GA" "RI"     "VA"     "DE"     "MD"     "NJ"     "CT"    
[17] "MA"     "NY"     "LA/MS"  "AL/FLW"
```

```r
lis <- c("TX", "LA", "LA/MS", "MS", "AL", "AL/FLW")  # filter out the Gulf states
d1 <- d[-which(d$NEW_STA %in% lis), ]
unique(d1$NEW_STA)
```

```
 [1] "NC"     "FLE"    "FLW"    "SC"     "GA"     "FLE/GA" "RI"     "VA"    
 [9] "DE"     "MD"     "NJ"     "CT"     "MA"     "NY"    
```

```r
table(d1$FL_REG, d1$NEW_STA, useNA = "always")
```

```
         CT   DE  FLE FLE/GA  FLW   GA   MA   MD   NC   NJ   NY   RI   SC   VA
  1       0    0    0      0  387    0    0    0    0    0    0    0    0    0
  2       0    0    0      0  110    0    0    0    0    0    0    0    0    0
  3       0    0    0      0  568    0    0    0    0    0    0    0    0    0
  4       0    0 1022      0    0    0    0    0    0    0    0    0    0    0
  5       0    0  312      0    0    0    0    0    0    0    0    0    0    0
  <NA>   14  114    0   1794  174   54   20  125 1344  108   73   62  644  165

       <NA>
```

```
  1        0
  2        0
  3        0
  4        0
  5        0
  <NA>     0
```

```
dim(d1)
```

```
[1] 7090   83
```

```
d1 <- d1[-which(d1$FL_REG == 1 | d1$FL_REG == 2), ]   # filter out FL regions 1 and 2 but r
dim(d1)
```

```
[1] 6593   83
```

```
d1 <- d1[-which(is.na(d1$FL_REG) & d1$NEW_STA == "FLW"), ]  # filter out FLW with unknown F
dim(d1)
```

```
[1] 6419   83
```

```
table(d1$FL_REG, d1$NEW_STA, useNA = "always")  # retained are all Atlantic states plus reg
```

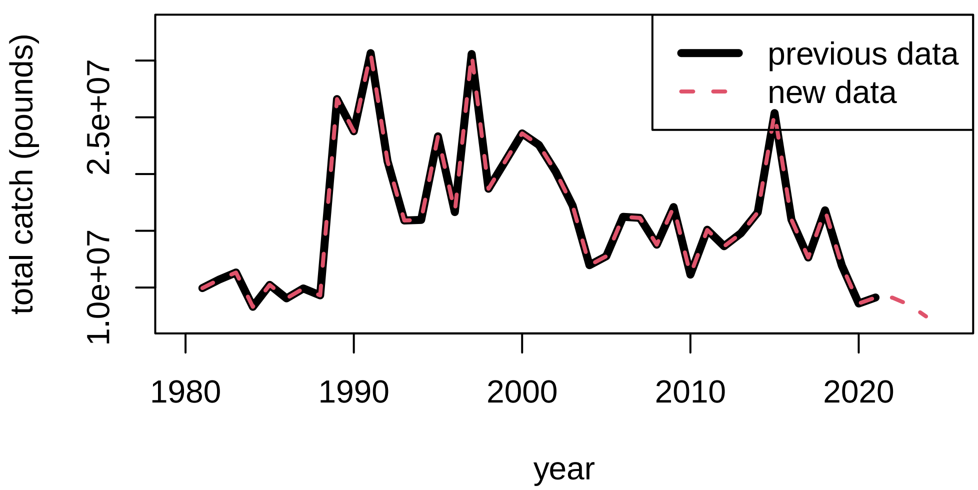
```
         CT    DE   FLE FLE/GA  FLW    GA    MA    MD    NC    NJ    NY    RI    SC    VA
  3       0     0     0      0  568     0     0     0     0     0     0     0     0     0
  4       0     0  1022      0    0     0     0     0     0     0     0     0     0     0
  5       0     0   312      0    0     0     0     0     0     0     0     0     0     0
  <NA>   14   114     0   1794    0    54    20   125  1344   108    73    62   644   165

       <NA>
  3       0
  4       0
  5       0
  <NA>    0
```

Now we can summarize the total dolphin landings in weight for the U.S. Atlantic, and compare to previous values sent for quota monitoring. We see that the values are almost identical.

```
# summarize by year and compare --------------------------

tot1 <- tapply(d1$LBSEST_SECWWT, d1$YEAR, sum, na.rm = T)
head(tot1)
```

```
    1981     1982     1983     1984     1985     1986
 9952644 10701659 11325575  8303073 10236339  9047438
```

```
old <- read.csv("data/RecreationalDolphinLandings_SAandGOM_MLarkin.csv", skip = 6)
plot(old$Year, old$ATL, type = "l", xlab = "year", ylab = "total catch (pounds)", lwd = 4,
     xlim = c(1980, 2025), ylim = c(7*10^6, 3.3*10^7))
lines(as.numeric(names(tot1)), tot1, lwd = 2, col = 2, lty = 2)
legend("topright", c("previous data", "new data"), col = c(1, 2), lwd = c(4, 2), lty = c(1,
```



2.3 Partition catches by area and sector

Now we will partition out the total catches according to the dolphin MSE areas. The areas are as follows:

- FLK: Florida Keys to Indian River County County (FL)
- NCFL: Brevard (FL) to Southern NC (South of Hatteras)
- NNC: Northern NC (North of Hatteras to NC/VA border)
- VBM: Virginia to Maine

```
# separate out 4 regions -------------------------

d1$region <- "VBM"

# select for FLK
# FL_REG codes: 4: SE FL- Miami-Dade to Indian River County.; 5: NE FL- Brevard to Nassau C
table(d1$FL_REG, d1$NEW_STA, useNA = "always")
```

```
         CT   DE  FLE FLE/GA  FLW   GA   MA   MD   NC   NJ   NY   RI   SC   VA
  3       0    0    0      0  568    0    0    0    0    0    0    0    0    0
  4       0    0 1022      0    0    0    0    0    0    0    0    0    0    0
  5       0    0  312      0    0    0    0    0    0    0    0    0    0    0
  <NA>   14  114    0   1794    0   54   20  125 1344  108   73   62  644  165

       <NA>
  3       0
  4       0
  5       0
  <NA>    0
```

```
d1$region[which(d1$FL_REG == 3 | d1$FL_REG == 4)] <- "FLK"

# select for NCFL
d1$region[which(d1$FL_REG == 5)] <- "NCFL"
lis <- c("SC", "GA", "FLE/GA")           # NCFL states
d1$region[which(d1$NEW_STA %in% lis)] <- "NCFL"
# NC_REG codes: N: North of Cape Hatteras; S: South of Cape Hatteras
table(d1$NC_REG, d1$NEW_STA, useNA = "always")
```

```
          CT    DE   FLE FLE/GA   FLW    GA    MA    MD    NC    NJ    NY    RI    SC    VA
          14   114  1334   1794   568    54    20   125   586   108    73    62   644   165
  N        0     0     0      0     0     0     0     0   420     0     0     0     0     0
  S        0     0     0      0     0     0     0     0   338     0     0     0     0     0
  <NA>     0     0     0      0     0     0     0     0     0     0     0     0     0     0

        <NA>
           0
  N        0
  S        0
  <NA>     0
```

```
d1$region[which(d1$NC == "S")] <- "NCFL"

# select for NNC
d1$region[which(d1$NC == "N")] <- "NNC"

table(d1$FL_REG, d1$NEW_STA, d1$region)
```

```
, ,  = FLK


      CT    DE   FLE FLE/GA   FLW    GA    MA    MD    NC    NJ    NY    RI    SC    VA
  3    0     0     0      0   568     0     0     0     0     0     0     0     0     0
  4    0     0  1022      0     0     0     0     0     0     0     0     0     0     0
  5    0     0     0      0     0     0     0     0     0     0     0     0     0     0

, ,  = NCFL


      CT    DE   FLE FLE/GA   FLW    GA    MA    MD    NC    NJ    NY    RI    SC    VA
  3    0     0     0      0     0     0     0     0     0     0     0     0     0     0
  4    0     0     0      0     0     0     0     0     0     0     0     0     0     0
  5    0     0   312      0     0     0     0     0     0     0     0     0     0     0

, ,  = NNC


      CT    DE   FLE FLE/GA   FLW    GA    MA    MD    NC    NJ    NY    RI    SC    VA
```

```
  3   0   0   0     0   0   0   0   0   0   0   0   0   0   0
  4   0   0   0     0   0   0   0   0   0   0   0   0   0   0
  5   0   0   0     0   0   0   0   0   0   0   0   0   0   0

, ,  = VBM


     CT   DE  FLE FLE/GA  FLW   GA   MA   MD   NC   NJ   NY   RI   SC   VA
  3   0    0    0      0    0    0    0    0    0    0    0    0    0    0
  4   0    0    0      0    0    0    0    0    0    0    0    0    0    0
  5   0    0    0      0    0    0    0    0    0    0    0    0    0    0
```

```r
table(d1$NC_REG, d1$NEW_STA, d1$region)  # check classification
```

```
, ,  = FLK


     CT   DE  FLE FLE/GA  FLW   GA   MA   MD   NC   NJ   NY   RI   SC   VA
      0    0 1022      0  568    0    0    0    0    0    0    0    0    0
  N   0    0    0      0    0    0    0    0    0    0    0    0    0    0
  S   0    0    0      0    0    0    0    0    0    0    0    0    0    0

, ,  = NCFL


     CT   DE  FLE FLE/GA  FLW   GA   MA   MD   NC   NJ   NY   RI   SC   VA
      0    0  312   1794    0   54    0    0    0    0    0    0  644    0
  N   0    0    0      0    0    0    0    0    0    0    0    0    0    0
  S   0    0    0      0    0    0    0    0  338    0    0    0    0    0

, ,  = NNC


     CT   DE  FLE FLE/GA  FLW   GA   MA   MD   NC   NJ   NY   RI   SC   VA
      0    0    0      0    0    0    0    0    0    0    0    0    0    0
  N   0    0    0      0    0    0    0    0  420    0    0    0    0    0
  S   0    0    0      0    0    0    0    0    0    0    0    0    0    0

, ,  = VBM
```

```
      CT  DE  FLE FLE/GA FLW  GA  MA  MD  NC  NJ  NY  RI  SC  VA
      14 114    0      0   0   0  20 125 586 108  73  62   0 165
  N    0   0    0      0   0   0   0   0   0   0   0   0   0   0
  S    0   0    0      0   0   0   0   0   0   0   0   0   0   0
```

Now we will separate out the catches by recreational sector and plot the totals by region and sector.

```
# separate out by for hire vs. private rec -------------------------
d1$sector <- "ForHire"

d1$sector[which(d1$NEW_MODEN == "Shore")] <- "Rec"
d1$sector[which(d1$NEW_MODEN == "Priv")]  <- "Rec"

table(d1$sector, d1$NEW_MODEN, useNA = "always")    # check classification
```

```
          Cbt Cbt/Hbt  Hbt Priv Shore <NA>
  ForHire 1616     119 2912    0     0    0
  Rec        0       0    0 1750    22    0
  <NA>       0       0    0    0     0    0
```

```
tot <- tapply(d1$LBSEST_SECWWT, list(d1$YEAR, d1$region, d1$sector), sum, na.rm = T)

head(tot)
```

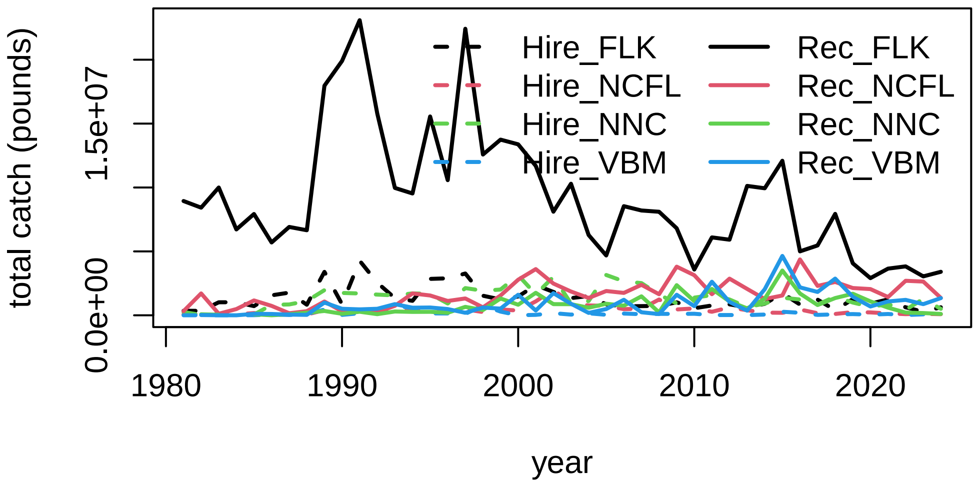
```
, , ForHire

           FLK      NCFL       NNC       VBM
1981  354639.5  74702.50 256755.24  1400.247
1982  347458.2  92271.42  27578.44  2749.224
1983 1020005.8  41784.04  88690.53  5379.177
1984 1021086.8  74658.81        NA  4203.048
1985  717523.0 172989.65  58514.13  8880.857
1986 1562354.0  63555.89 817192.26 61658.193

, , Rec
```

```
          FLK      NCFL       NNC        VBM
1981  8942641  322505.7        NA         NA
1982  8422673 1711834.7 78367.161  18727.290
1983 10000662  119370.4 44226.032   5456.237
1984  6717044  486079.7        NA      0.000
1985  7925301 1162708.4 77477.096 112944.274
1986  5699311  736872.0  2549.586 103945.134
```

```
fin <- data.frame(cbind(tot[, , 1], tot[, , 2]))
names(fin) <- c("Hire_FLK", "Hire_NCFL", "Hire_NNC", "Hire_VBM", "Rec_FLK", "Rec_NCFL", "Re
head(fin)
```

```
      Hire_FLK Hire_NCFL  Hire_NNC  Hire_VBM  Rec_FLK  Rec_NCFL   Rec_NNC
1981  354639.5  74702.50 256755.24  1400.247  8942641  322505.7        NA
1982  347458.2  92271.42  27578.44  2749.224  8422673 1711834.7 78367.161
1983 1020005.8  41784.04  88690.53  5379.177 10000662  119370.4 44226.032
1984 1021086.8  74658.81        NA  4203.048  6717044  486079.7        NA
1985  717523.0 172989.65  58514.13  8880.857  7925301 1162708.4 77477.096
1986 1562354.0  63555.89 817192.26 61658.193  5699311  736872.0  2549.586
        Rec_VBM
1981         NA
1982  18727.290
1983   5456.237
1984      0.000
1985 112944.274
1986 103945.134
```

```
matplot(rownames(fin), fin, lty = c(rep(2, 4), rep(1, 4)), col = rep(1:4, 2), type = "l", l
        xlab = "year", ylab = "total catch (pounds)")
legend("topright", colnames(fin), lty = c(rep(2, 4), rep(1, 4)), col = rep(1:4, 2), lwd = 2
```




```r
# check that totals match
cor(rowSums(fin, na.rm = T), tot1)
```

```
[1] 1
```

```r
mean(rowSums(fin, na.rm = T) - tot1)   # very tiny differences
```

```
[1] 4.233284e-11
```

Here are the catches (in pounds) for years 1981 to 2024.

```r
fin
```

```
      Hire_FLK  Hire_NCFL   Hire_NNC   Hire_VBM  Rec_FLK  Rec_NCFL     Rec_NNC
1981  354639.5   74702.50  256755.24   1400.247  8942641  322505.7          NA
1982  347458.2   92271.42   27578.44   2749.224  8422673 1711834.7   78367.161
1983 1020005.8   41784.04   88690.53   5379.177 10000662  119370.4   44226.032
1984 1021086.8   74658.81         NA   4203.048  6717044  486079.7          NA
1985  717523.0  172989.65   58514.13   8880.857  7925301 1162708.4   77477.096
```

```
1986 1562354.0   63555.89  817192.26  61658.193  5699311  736872.0    2549.586
1987 1774558.7   61243.58  848994.25  24967.877  6913585  161627.1   79540.365
1988  826480.6   62392.87 1114627.05  94676.212  6655923  321023.1  202768.950
1989 3398703.5  405140.01 1970776.23 451477.768 17965047 1079951.5  339371.567
1990  868572.6  277783.17 1749965.55  31079.442 19900865  312333.3  118539.960
1991 4268773.7  225552.64 1706751.50 159172.855 23093044  457667.1  277389.646
1992 2548407.4  189514.34 1619685.06 161149.703 15788853  245395.8   89160.860
1993 1343093.6  391254.33 1555850.09 741955.369  9967356  742741.8  296399.199
1994 1131831.7  437642.72 1713981.98 573551.716  9532763 1708942.1  264907.193
1995 2848691.7  667475.30 1673435.48 145798.587 15558489 1550136.0  271442.073
1996 2892436.3  309604.42  923128.81 141976.431 10574454 1121937.8  196579.537
1997 3271271.6  466891.67 2129471.46 110115.485 22423256 1316502.7  660257.843
1998 1532460.0  252620.73 1921834.08 779364.855 12578694  619824.1  412304.292
1999 1238462.0  457880.10 2014752.29 288625.130 13745135 1548217.4 1314805.374
2000 1471481.3  369553.33 3139555.98   3016.589 13379647 2784424.2  826980.337
2001 2361751.6 1090421.00 1638707.54  34335.241 11690170 3614370.3 1760721.144
2002 1828918.7 1993778.99 3005172.92 168406.810  8105262 2483946.6  871930.660
2003 1342339.1  958997.12  928894.01  55690.419 10294918 1860435.4  868200.254
2004 1509660.5  771820.65 1116373.11 155510.796  6281794 1354303.7  593257.716
2005  858090.6  734896.20 3153480.89  38020.553  4681210 1898945.9  904360.210
2006  700089.5  483337.12 2656240.22 136715.477  8540432 1751853.5  749541.413
2007  718651.9  506776.30 2518430.49  84908.300  8200488 2383417.1 1483630.590
2008  736213.2 1235299.71 1620865.87 113592.877  8100550 1648662.2  223565.185
2009 1016583.6  458562.82  928419.86 106262.915  6809447 3805127.7 2359177.817
2010  552080.9  534671.06 1384434.93 120050.410  3579591 3139199.6 1110479.730
2011  768597.7  275546.65 1577019.96  17844.729  6097050 1651096.4 2063315.642
2012  856261.4  615430.34 1236410.76  21887.318  5916144 2867599.1 1082607.063
2013  614150.1  394570.11  640465.60  16064.763 10124799 2064256.6  540676.961
2014  897508.4  210198.76  916695.96  56760.758  9940020 1272477.3 1289704.416
2015 1729825.9  196326.97 1374908.91 277867.986 12093928 1548428.5 3505194.743
2016  841662.5  454053.51 1236565.76 195837.854  5002273 4372622.6 1704661.669
2017 1227827.8  161188.31  835080.86  31185.061  5464893 2297930.0  801324.151
2018  401703.0  105328.49 1465087.58  72211.785  7935729 2606091.5 1347724.203
2019 1261827.1  250624.40  998102.45  93718.699  4064867 2132483.1 1687239.863
2020  910431.4  225493.79  667501.79  51621.152  2906913 2052222.0 1089245.171
2021 1244842.2  146893.15  871760.94 100997.979  3652596 1428508.3  596672.480
2022  627020.6   90504.71  427953.34  18514.133  3828113 2701619.8  211394.537
2023  274532.5  118528.80 1281254.55  72728.020  3045801 2639445.1  175189.224
2024  629327.2   85572.19  506558.07  18752.239  3404227 1358519.1  106307.278
         Rec_VBM
1981          NA
```

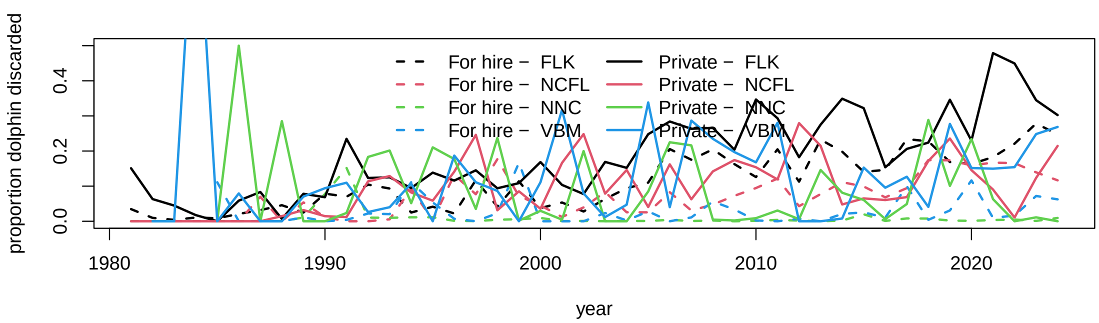
```
1982   18727.290
1983    5456.237
1984       0.000
1985  112944.274
1986  103945.134
1987   52139.772
1988   33566.163
1989  996945.390
1990  509424.981
1991  464786.733
1992  509297.261
1993  871948.601
1994  594467.414
1995  609302.619
1996  486106.187
1997  197871.963
1998  606769.298
1999  523623.688
2000 1608478.920
2001  372623.273
2002 1732356.036
2003  904780.558
2004  186538.567
2005  489236.451
2006 1214497.040
2007  236769.123
2008   96818.461
2009 1607919.004
2010  717410.858
2011 2626410.052
2012 1041694.282
2013  364492.286
2014 2040329.550
2015 4645796.249
2016 2186126.672
2017 1826822.849
2018 2871124.410
2019 1440420.940
2020  689099.471
2021 1083268.811
2022 1206278.324
```

```
2023  872490.886
2024 1346606.842
```

2.4 Looking at extent of discarding

Discard estimates (known as B2 estimates) are provided in numbers by the MRIP survey but are not available in the Southeast Region Headboat Survey. We can look at the overall numbers of dolphinfish caught versus reported released from the MRIP survey to get an understanding of the extent of discarding.

```
d1$perdis <- d1$B2 / (d1$AB1 + d1$B2)  # percent discarded is B2 (fish released alive) / to

# calculate mean discard rate by year, region, sector ----------------
dis <- tapply(d1$perdis, list(d1$YEAR, d1$region, d1$sector), mean, na.rm = T)
disr <- tapply(d1$perdis, list(d1$region, d1$sector), mean, na.rm = T)

matplot(rownames(dis), cbind(dis[, , 1], dis[, , 2]), lty = c(rep(2, 4), rep(1, 4)), col =
        type = "l", lwd = 2, xlab = "year", ylab = "proportion dolphin discarded", ylim = c
legend("top", c(paste("For hire - ", colnames(dis)), paste("Private - ", colnames(dis))),
       lty = c(rep(2, 4), rep(1, 4)), col = rep(1:4, 2), lwd = 2, ncol = 2, bty = "n")
```




```
round(disr * 100, 2)  # discarding rates by region and sector
```

```
     ForHire   Rec
FLK    10.79 18.76
NCFL    9.44 11.23
```

```
NNC      0.94  9.62
VBM      2.94 12.69
```

According to the MRIP survey and the reported numbers, discarding generally increases over time, excluding some highly anomalous estimates from the early years which are driven by low sample sizes. Discarding is generally higher in the private sector compared to the for-hire sector, and is highest in the Florida Keys region.

2.5 Split the catches by season

The dolphin MSE operates on a seasonal time step and thus the catches and discards must by parsed apart by individual season and year. This is not straightforward because MRIP estimates are reported in two-month waves (Jan - Feb, Mar - Apr, May - Jun, etc.), and the seasons are 3-month periods (Dec - Feb, Mar - May, Jun - Aug, Sep - Nov). For the two waves that have to be split into seasons (e.g., December is lumped with January and February; June is lumped with July and August), we split the wave 50/50%, i.e, we make the assumption that half of the catch in that wave occurs in one season and the other half occurs in the next season.

```
# define seasons ------------------------------
table(d1$WAVE, useNA = "always")
```

```
   0    1    2    3    4    5    6 <NA>
   5  407  718 1230 1439 1095  581  944
```

```
d1$YRWAVE <- paste0(d1$YEAR, "_", d1$WAVE)

totw <- tapply(d1$LBSEST_SECWWT, list(d1$YRWAVE, d1$region, d1$sector), sum, na.rm = T)

f <- data.frame(cbind(totw[, , 1], totw[, , 2]))
names(f) <- c("Hire_FLK", "Hire_NCFL", "Hire_NNC", "Hire_VBM", "Rec_FLK", "Rec_NCFL", "Rec_
head(f)
```

```
         Hire_FLK Hire_NCFL   Hire_NNC Hire_VBM   Rec_FLK  Rec_NCFL Rec_NNC
1981_2         NA        NA         NA       NA 2793285.5 270968.46      NA
1981_3 300873.764        NA 254049.420       NA 3915698.3        NA      NA
```

```
1981_4   50017.460        NA   2705.821        NA  905621.2  51537.26        NA
1981_5    3748.301        NA         NA        NA  479834.0        NA        NA
1981_6          NA        NA         NA        NA  848201.5        NA        NA
1981_NA         NA   74702.5         NA  1400.247        NA        NA        NA
        Rec_VBM
1981_2       NA
1981_3       NA
1981_4       NA
1981_5       NA
1981_6       NA
1981_NA      NA
```

```
f$YEAR <- as.numeric(substr(rownames(f), 1, 4))
f$WAVE <- as.numeric(substr(rownames(f), 6, 8))
head(f)   # table with catches separated by wave and year
```

```
          Hire_FLK Hire_NCFL   Hire_NNC Hire_VBM   Rec_FLK   Rec_NCFL Rec_NNC
1981_2          NA        NA         NA       NA 2793285.5 270968.46      NA
1981_3  300873.764        NA 254049.420       NA 3915698.3        NA      NA
1981_4   50017.460        NA   2705.821       NA  905621.2  51537.26      NA
1981_5    3748.301        NA         NA       NA  479834.0        NA      NA
1981_6          NA        NA         NA       NA  848201.5        NA      NA
1981_NA         NA   74702.5         NA 1400.247        NA        NA      NA
        Rec_VBM YEAR WAVE
1981_2       NA 1981    2
1981_3       NA 1981    3
1981_4       NA 1981    4
1981_5       NA 1981    5
1981_6       NA 1981    6
1981_NA      NA 1981   NA
```

```
f[is.na(f)] <- 0

f <- f[which(f$YEAR >= 1985), ]
f <- f[which(f$YEAR <= 2024), ]

table(f$YEAR)
```

```
1985 1986 1987 1988 1989 1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000
   7    7    7    7    7    7    7    7    7    7    7    7    6    6    6    6
2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016
   6    6    6    6    6    6    6    6    6    6    6    6    7    7    7    6
2017 2018 2019 2020 2021 2022 2023 2024
   6    6    6    6    6    6    6    6
```

```
table(f$WAVE)
```

```
 0  1  2  3  4  5  6
15 40 40 40 40 40 40
```

```
# deal with NAs in many locations ---------------
# calculate percentages of catch by wave, for each region and fleet
avw <- tapply(d1$LBSEST_SECWWT, list(d1$WAVE, d1$region, d1$sector), sum, na.rm = T)
avw <- avw[-1, , ]
avw <- data.frame(cbind(avw[, , 1], avw[, , 2]))
names(avw) <- c("Hire_FLK", "Hire_NCFL", "Hire_NNC", "Hire_VBM", "Rec_FLK", "Rec_NCFL", "Re
avwp <- apply(avw, 2, function (x) x / sum(x, na.rm = T))
#colSums(avwp)

# go through each column, find NAs and fill in waves by percentages
pre_colsum <- colSums(f, na.rm = T)  # column sums to check later

lis <- which(f$WAVE == 0)  # list of rows with NA wave data

for (i in 1:8) {       # loop through each column
  for(j in lis) {      # loop through list of NA waves
    if (f[j, i] > 0) {  # if there are NA wave catches, split by known proportion from othe
      miscatch <- avwp[, i] * f[j, i]    # missing catch is known proportion * NA wave catch
      f[which(f$YEAR == f$YEAR[j] & f$WAVE != 0), i] <- f[which(f$YEAR == f$YEAR[j] & f$WAV
      f[j, i] <- 0   # set to zero after catch is attributed to waves
    }
  }
}
f[lis, 1:8] # should be all zeros now
```

```
        Hire_FLK Hire_NCFL Hire_NNC Hire_VBM Rec_FLK Rec_NCFL Rec_NNC Rec_VBM
```

```
1985_NA        0        0        0        0        0        0        0        0
1986_NA        0        0        0        0        0        0        0        0
1987_NA        0        0        0        0        0        0        0        0
1988_NA        0        0        0        0        0        0        0        0
1989_NA        0        0        0        0        0        0        0        0
1990_NA        0        0        0        0        0        0        0        0
1991_NA        0        0        0        0        0        0        0        0
1992_NA        0        0        0        0        0        0        0        0
1993_NA        0        0        0        0        0        0        0        0
1994_NA        0        0        0        0        0        0        0        0
1995_NA        0        0        0        0        0        0        0        0
1996_NA        0        0        0        0        0        0        0        0
2013_0         0        0        0        0        0        0        0        0
2014_0         0        0        0        0        0        0        0        0
2015_0         0        0        0        0        0        0        0        0
```

```
pos_colsum <- colSums(f, na.rm = T)
pre_colsum[1:8] == pos_colsum[1:8]  # check that catch was parsed correctly - column sums s
```

```
 Hire_FLK Hire_NCFL  Hire_NNC  Hire_VBM   Rec_FLK  Rec_NCFL   Rec_NNC   Rec_VBM
     TRUE      TRUE      TRUE      TRUE      TRUE      TRUE      TRUE      TRUE
```

```
f <- f[which(f$WAVE != 0), ]  # remove NA waves which now contain no catch

# split waves 3 and 6 50/50% to parse into seasons ------------------
lis <- which(f$WAVE == 3 | f$WAVE == 6)  # list of 3 and 6 waves for all years

for (i in lis) {          # go through list of waves
  temp <- f[i, ] / 2      # split catch in half
  temp[9] <- f$YEAR[i]    # reformat year and wave data columns
  temp[10] <- f$WAVE[i]
  temp <- rbind(temp, temp)   # create two half-waves and relabel with new wave reference n
  temp[1, 10] <- temp[1, 10] - 0.5
  temp[2, 10] <- temp[2, 10] + 0.5
  f <- rbind(f, temp)         # concatenate with main data table
  }

f <- f[-lis, ]  # remove waves that have been split in half
```

```
pos_colsum <- colSums(f, na.rm = T)
pre_colsum[1:8] == pos_colsum[1:8]  # check that catch was parsed correctly - column sums s
```

```
 Hire_FLK Hire_NCFL  Hire_NNC  Hire_VBM   Rec_FLK  Rec_NCFL   Rec_NNC   Rec_VBM
     TRUE      TRUE      TRUE      TRUE      TRUE      TRUE      TRUE      TRUE
```

```
f <- f[order(f$YEAR, f$WAVE), ]  # reorder table in chronological order

f$YEAR[which(f$WAVE == 6.5)] <- f$YEAR[which(f$WAVE == 6.5)] + 1  # add year to December wa
f$WAVE[which(f$WAVE == 6.5)] <- 0.5  # change December wave to small number so it gets summ

table(f$WAVE)
```

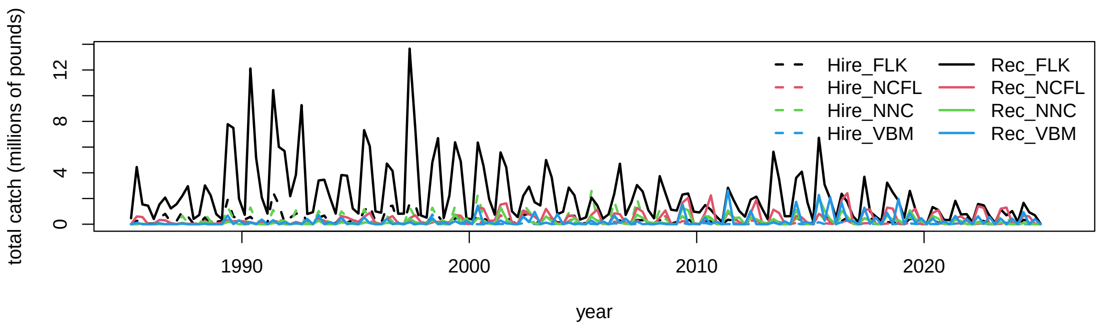
```
0.5   1   2 2.5 3.5   4   5 5.5
 40  40  40  40  40  40  40  40
```

```
f$seas <- cut(f$WAVE, breaks = c(0, 1.5, 3, 4.5, 6))
#table(f$seas)
f$seas1 <- NA
f$seas1[which(f$seas == "(0,1.5]")] <- "DJF"  # Dec Jan Feb is 0.5 and 1
f$seas1[which(f$seas == "(1.5,3]")] <- "MAM"  # Mar Apr May is 2 and 2.5
f$seas1[which(f$seas == "(3,4.5]")] <- "JJA"  # Jun Jul Aug is 3.5 and 4
f$seas1[which(f$seas == "(4.5,6]")] <- "SON"  # Sep Oct Nov is 5 and 5.5

temp <- tapply(f[, 1], list(f$seas1, f$YEAR), sum, na.rm = T)
temp1 <- matrix(temp)
for (i in 2:8) {
  temp2 <- matrix(tapply(f[, i], list(f$seas1, f$YEAR), sum, na.rm = T))
  temp1 <- cbind(temp1, temp2)
  }

findat <- data.frame(sort(rep(as.numeric(colnames(temp)), 4)),
                     rep(rownames(temp), ncol(temp)), temp1)
names(findat) <- c("year", "season", names(f)[1:8])

colSums(f[, 1:8], na.rm = T) == colSums(findat[, 3:10], na.rm = T)  # check that total catc
```

```
 Hire_FLK Hire_NCFL  Hire_NNC  Hire_VBM   Rec_FLK  Rec_NCFL   Rec_NNC   Rec_VBM
     TRUE      TRUE      TRUE      TRUE      TRUE      TRUE      TRUE      TRUE
```

We plot the results by region and sector, over the time series of year and season. We can clearly see the seasonality in the catch by region.

```
findat$yrseas <- findat$year + as.numeric(as.factor(findat$season))/4-0.125
matplot(findat$yrseas, findat[, 3:10]/10^6, lty = c(rep(2, 4), rep(1, 4)), col = rep(1:4, 2
        xlab = "year", ylab = "total catch (millions of pounds)")
legend("topright", colnames(findat[, 3:10]), lty = c(rep(2, 4), rep(1, 4)), col = rep(1:4,
```




```
# check that totals match
cbind(colSums(findat[, 3:10], na.rm = T), colSums(fin[-c(1:5), ], na.rm = T))
```

```
               [,1]      [,2]
Hire_FLK   55604999  54887476
Hire_NCFL  16909916  16736926
Hire_NNC   57944903  57886388
Hire_VBM    5836953   5828072
Rec_FLK   371462933 363537632
Rec_NCFL   70803866  69641157
Rec_NNC    32556596  32479119
Rec_VBM    40081529  39968585
```

```
# the values should be close but slightly >1 because the last data set includes December 19
colSums(findat[, 3:10], na.rm = T) / colSums(fin[-c(1:5), ], na.rm = T)
```

```
 Hire_FLK Hire_NCFL  Hire_NNC  Hire_VBM   Rec_FLK  Rec_NCFL   Rec_NNC   Rec_VBM
 1.013073  1.010336  1.001011  1.001524  1.021800  1.016696  1.002385  1.002826
```

```
write.csv(findat, file = "data/MRIP_catch_by_season_4_areas.csv", row.names = FALSE)
```

3 Analysis of international landings estimates from Sea Around Us database

Here we will explore commercial and recreational landings data as estimated by the Sea Around Us Project. This data set includes reported data for fisheries worldwide as well as reconstructed estimates for unreported landings. For each country, project scientists assemble available data on landings and use additional sociological and fishing data plus expert information and opinion, to produce their best estimates of total landings from all fishing sectors and for all species globally.

3.1 Data access and upload

For catches within each country's exclusive economic zones, data are downloaded download data from the Sea Around Us site(Pauly, Zeller, and Palomares, 2020.). In the Biodiversity by Taxon Tools and Data, we filter by EEZ for All regions, and then filter by Taxon at the Species level for Common dolphinfish (Coryphaena hippurus). After clicking "View graphs" one can access the full data set using the "Download Data" button. The high seas data are accessed in the Basic Search High seas section of the site. The individual regions can be clicked on (e.g., Atlantic, Western Central) and then the data are dowloaded using the Download Data button.

The current version of data in use is version 50.1, accessed June 2024 (the data files can be found on github in the SEFSC-dolphin-analyses/data /SAU/ folder). This was still the most recent version of data available as of June 2025.

3.2 Read in the high seas landings

First we read in the high seas landings for the Western Central Atlantic and Northwest Atlantic zones. These are equivalent to FAO areas 21 and 31.

```
# clear workspace
rm(list = ls())

if(!require("pals")) install.packages("pals")
library(pals)

# find data files
hslis <- dir("data/SAU/")[grep("HighSeas", dir("data/SAU/"))]

# identify areas 21 and 31
hslis[1]; hslis[3]
```

```
[1] "SAU HighSeas 21 v50-1.csv"
```

```
[1] "SAU HighSeas 31 v50-1.csv"
```

```
#download data
h1 <- read.csv(paste0("data/SAU/", hslis[1]))
h3 <- read.csv(paste0("data/SAU/", hslis[3]))

# confirm correct areas
#unique(h1$area_name)
#unique(h3$area_name)

hs <- rbind(h1, h3)
```

We extract the data for dolphinfish and look at the characteristics of the landings. Most of the landings are in the Western Central Atlantic, and the primary fishing entities are Venezuela and Saint Lucia. The fishing sector is all industrial, and the vast majority of the catch is reported landings, with only a tiny fraction of discards or unreported landings listed in the database. The gear type is primarily long line.

```
# check species name and filter out dolphinfish
unique(hs$common_name)[grep("dolphin", unique(hs$common_name))]
```

```
[1] "Common dolphinfish"
```

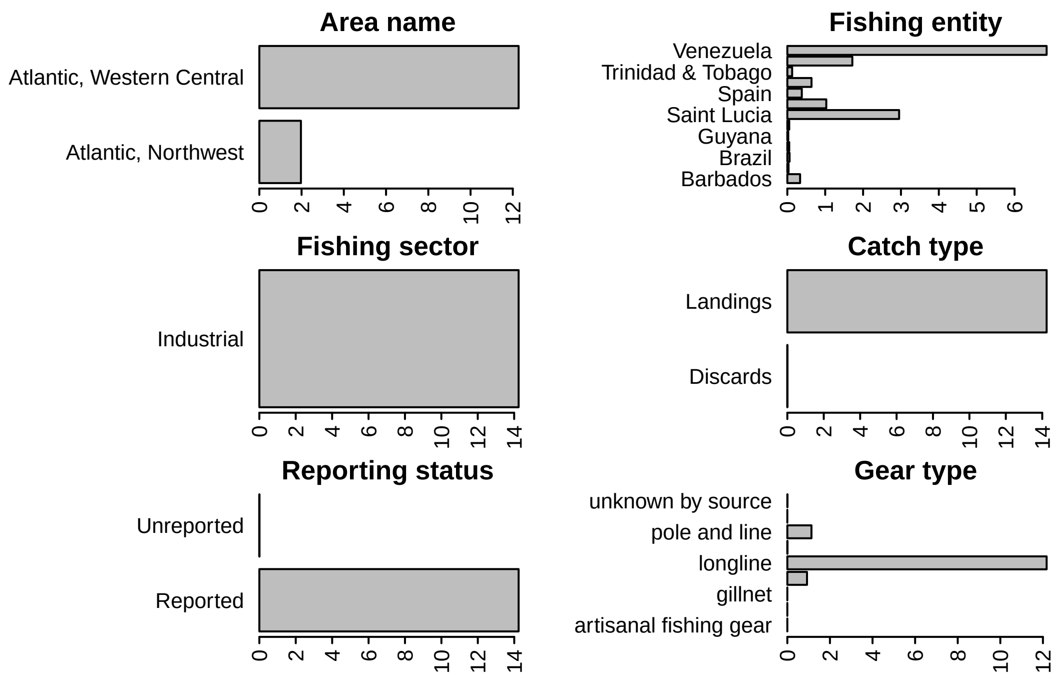
```r
hs <- hs[which(hs$common_name == "Common dolphinfish"), ]

# look at the categories within each column of data
apply(hs[1:13], 2, table)
```

```
$area_name

      Atlantic, Northwest Atlantic, Western Central
                      113                       160

$area_type

high_seas
      273

$year

1985 1986 1999 2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012
   1    1    1    1    1    1    1    6    4    1    5   10    5    8   10   12
2013 2014 2015 2016 2017 2018 2019
  24   22   30   33   33   29   34

$scientific_name

Coryphaena hippurus
                273

$common_name

Common dolphinfish
               273

$functional_group

Large pelagics (>=90 cm)
                     273

$commercial_group

Perch-likes
```

```
        273

$fishing_entity

                     Barbados                          Belize 
                            6                               3 
                       Brazil                          Canada 
                           19                              18 
                       Guyana                          Panama 
                            1                               1 
                  Saint Lucia Saint Vincent & the Grenadines 
                            5                              28 
                        Spain                        Suriname 
                           80                               1 
            Trinidad & Tobago                             USA 
                           14                              74 
                    Venezuela 
                           23 

$fishing_sector

Industrial 
       273 

$catch_type

Discards Landings 
       5      268 

$reporting_status

  Reported Unreported 
       253         20 

$gear_type

artisanal fishing gear           bottom trawl                gillnet 
                     7                      9                     11 
            hand lines               longline             mixed gear 
                    17                    187                      1 
         pole and line          pots or traps      unknown by source 
```

```
                   19                        9                          13

$end_use_type

                          Direct human consumption    Fishmeal and fish oil
                        5                      194                       37
                    Other
                       37
```

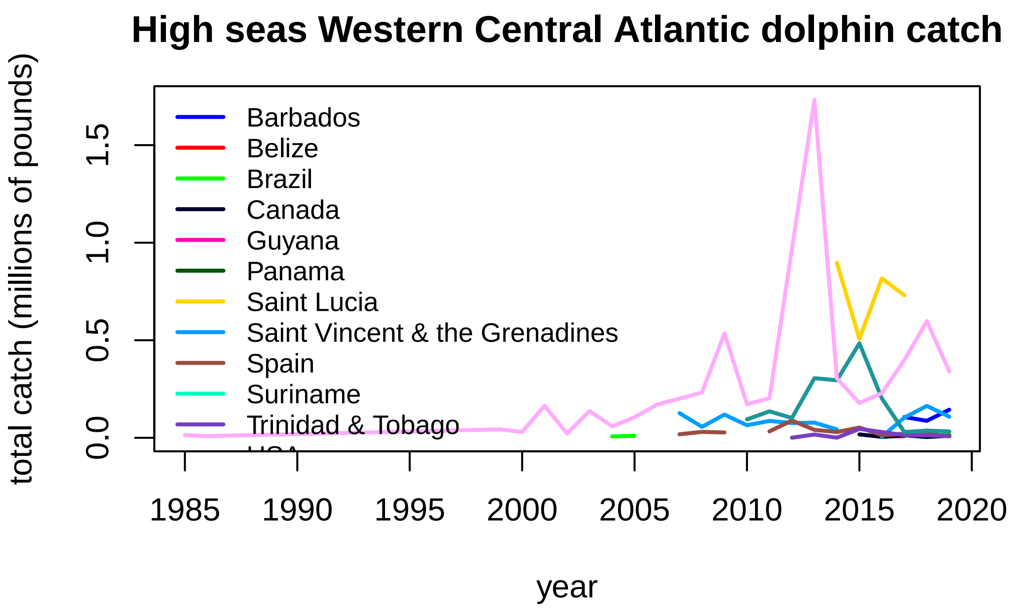
```
# convert from metric tonnes to pounds
hs$pounds <- hs$tonnes * 1000 * 2.20462

# produce barplot of key data fields
par(mar = c(3, 20, 3, 1), mfrow = c(3, 2), mex = 0.5)
barplot(tapply(hs$pounds/10^6, hs$area_name, sum, na.rm = T), las = 2, xlab = "millions of
#barplot(tapply(hs$pounds/10^6, hs$area_type, sum, na.rm = T), las = 2, ylab = "millions of
#barplot(tapply(hs$pounds/10^6, hs$year, sum, na.rm = T), las = 2, ylab = "millions of poun
barplot(tapply(hs$pounds/10^6, hs$fishing_entity, sum, na.rm = T), las = 2, xlab = "million
barplot(tapply(hs$pounds/10^6, hs$fishing_sector, sum, na.rm = T), las = 2, xlab = "million
barplot(tapply(hs$pounds/10^6, hs$catch_type, sum, na.rm = T), las = 2, xlab = "millions of
barplot(tapply(hs$pounds/10^6, hs$reporting_status, sum, na.rm = T), las = 2, xlab = "milli
barplot(tapply(hs$pounds/10^6, hs$gear_type, sum, na.rm = T), las = 2, xlab = "millions of
```



3.3 Summarize the high seas landings

We plot the annual catches by fishing entity to look at trends over time.

```
hs$fishing_entity <- as.factor(hs$fishing_entity)
hs$fishing_entity <- droplevels(hs$fishing_entity)

par(mar = c(5, 5, 3, 1), mfrow = c(1, 1))
tab <- tapply(hs$pounds, list(hs$year, hs$fishing_entity), sum, na.rm = T)
matplot(as.numeric(rownames(tab)), tab/10^6, type = "l", col = glasbey(ncol(tab)), lty = 1,
        main = "High seas Western Central Atlantic dolphin catch")
legend("topleft", colnames(tab), col = glasbey(ncol(tab)), lty = 1, lwd = 2, cex = 0.8, bty
```



Finally we output the subset of high seas dolphin catch for the two areas. Because we have already accounted for the USA landings in our national reporting and are aiming to only summarize international landings here, we filter out the USA landings from the Sea Around Us database.

```
hs <- hs[which(hs$fishing_entity != "USA"), ]

hs$year <- factor(hs$year, levels = 1950:2019)
tab <- tapply(hs$pounds, list(hs$year, hs$area_name), sum, na.rm = T)
tab[is.na(tab)] <- 0
tail(tab, 40)
```

```
     Atlantic, Northwest Atlantic, Western Central
1980                0.000                    0.000
1981                0.000                    0.000
1982                0.000                    0.000
1983                0.000                    0.000
1984                0.000                    0.000
1985                0.000                14037.129
1986                0.000                 8748.399
1987                0.000                    0.000
```

```
1988              0.000                0.000
1989              0.000                0.000
1990              0.000                0.000
1991              0.000                0.000
1992              0.000                0.000
1993              0.000                0.000
1994              0.000                0.000
1995              0.000                0.000
1996              0.000                0.000
1997              0.000                0.000
1998              0.000                0.000
1999              0.000            43168.296
2000              0.000            30284.627
2001              0.000           164272.223
2002              0.000            22147.477
2003              0.000           137103.181
2004              0.000            66058.258
2005              0.000           115171.863
2006              0.000           170522.101
2007          14687.527           330838.620
2008          28552.636           340863.719
2009          18719.121           662720.661
2010           4510.085           232093.837
2011          21589.424           300498.224
2012          72398.812          1059210.392
2013          34741.604          2471791.396
2014          27922.953          1247462.983
2015          90313.638           710059.125
2016          15319.433          1094893.891
2017          34806.933          1374259.739
2018          29328.358           885814.754
2019          21870.621           632747.157
```

```r
# write summary to merge with EEZ data
write.csv(tab, file = "data/SAU/high_seas_by_year.csv")
```

3.4 Read in the landings within excluzive economic zones (EEZs)

Now we read in the landings reported by each country or jurisdiction's respective EEZ. Because we have extracted dolphin landings for EEZs worldwide, we need to separate out the landings for the Atlantic Ocean basin.

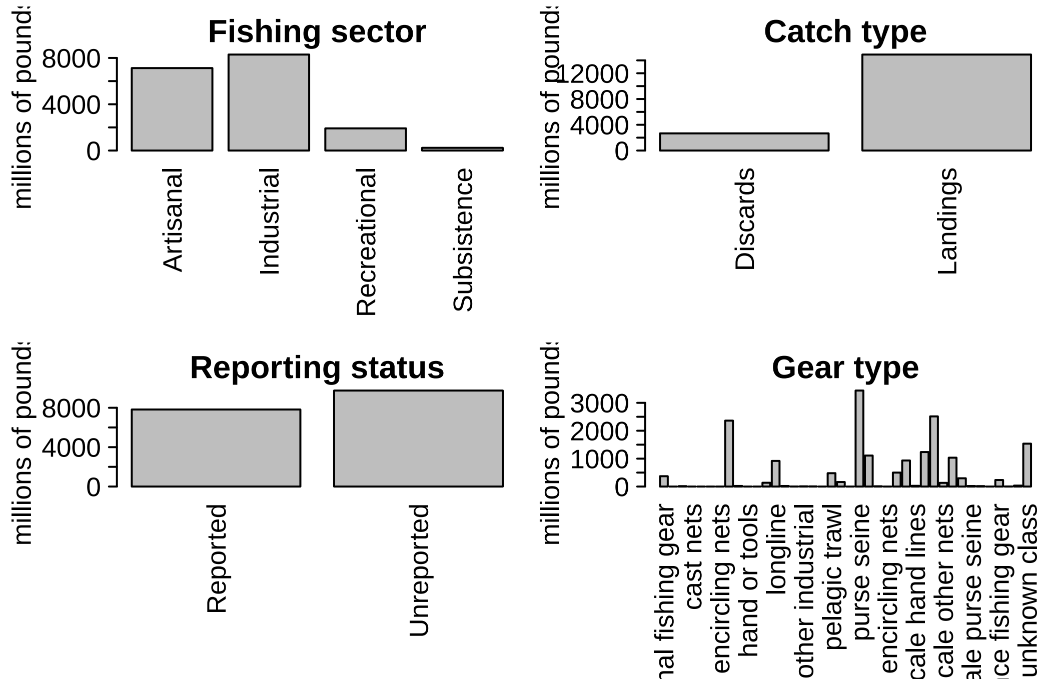
```
rm(list = ls())
d <- read.csv("data/SAU/SAU Taxa 600006 v50-1.csv", header = T, sep = ",")  # v. 50.1 - Jun
head(d)
```

```
     area_name area_type year     scientific_name        common_name
1    Australia       eez 1950 Coryphaena hippurus Common dolphinfish
2    Australia       eez 1950 Coryphaena hippurus Common dolphinfish
3    Australia       eez 1950 Coryphaena hippurus Common dolphinfish
4      Bahamas       eez 1950 Coryphaena hippurus Common dolphinfish
5     Barbados       eez 1950 Coryphaena hippurus Common dolphinfish
6 Bermuda (UK)       eez 1950 Coryphaena hippurus Common dolphinfish
          functional_group commercial_group fishing_entity fishing_sector
1 Large pelagics (>=90 cm)      Perch-likes      Australia     Industrial
2 Large pelagics (>=90 cm)      Perch-likes      Australia     Industrial
3 Large pelagics (>=90 cm)      Perch-likes      Australia     Industrial
4 Large pelagics (>=90 cm)      Perch-likes        Bahamas   Recreational
5 Large pelagics (>=90 cm)      Perch-likes       Barbados   Recreational
6 Large pelagics (>=90 cm)      Perch-likes   Bermuda (UK)      Artisanal
  catch_type reporting_status                 gear_type
1   Landings         Reported                  longline
2   Landings         Reported                  longline
3   Landings         Reported                  longline
4   Landings       Unreported recreational fishing gear
5   Landings       Unreported recreational fishing gear
6   Landings         Reported         small scale lines
              end_use_type       tonnes landed_value
1    Fishmeal and fish oil  0.003373202 3.438170e-05
2 Direct human consumption  6.739657104 1.040254e+04
3                    Other  0.003373202 3.438170e-05
4 Direct human consumption 48.665105143 8.262728e+00
5 Direct human consumption  0.009158559 4.846886e-04
6 Direct human consumption  1.511085514 3.199403e+03
```

```
#dim(d)
#apply(d[1:13], 2, table)

# convert tonnes to pounds
d$lbs <- d$tonnes * 1000 * 2.20462
```

We look at the characteristics of the landings within EEZs. These are largely industrial and artisanal in nature. Landings are the majority of the catch but discards are also present. There are more unreported landings than reported landings and a variety of gear types are used.

```
# look at characteristics of data
par(mfrow = c(2, 2), mar = c(12, 7, 3, 1), mex = 0.5, mgp = c(5, 1, 0))
barplot(tapply(d$lbs, d$fishing_sector, sum, na.rm = T)/10^6, las = 2, ylab = "millions of 
barplot(tapply(d$lbs, d$catch_type, sum, na.rm = T)/10^6, las = 2, ylab = "millions of poun
barplot(tapply(d$lbs, d$reporting_status, sum, na.rm = T)/10^6, las = 2, ylab = "millions o
barplot(tapply(d$lbs, d$gear_type, sum, na.rm = T)/10^6, las = 2, ylab = "millions of pound
```



3.5 Separating the EEZs by area

Now we have to manually assign the different EEZs to respective areas. We separate the geographical areas within the Atlantic Ocean (North and South) and lump all other EEZs into "other oceans." This latter category is largely the Pacific Ocean and to a lesser extent the Indian Ocean. We remove the high seas catch from this database as it is not ocean-specific and we have extracted the ocean-specific high seas catch previously.

```
# remove the high seas catch from this database
d <- d[which(d$area_type != "high_seas"), ]
d$fishing_entity <- as.factor(d$fishing_entity)
d$fishing_entity <- droplevels(d$fishing_entity)
table(d$area_type)  # this should be all "eez"
```

```
  eez
63056
```

```
# Western Central Atlantic EEZs
WCA <- c("Bahamas", "Barbados", "Cayman Isl. (UK)", "Dominica", "Dominican Republic", "Gren
         "Aruba (Netherlands)", "Haiti", "Guadeloupe  (France)", "Jamaica", "Martinique (Fr
         "Montserrat (UK)", "Nicaragua (Caribbean)", "Puerto Rico (USA)", "Saint Kitts & Ne
         "Saint Lucia", "Saint Vincent & the Grenadines", "Trinidad & Tobago", "US Virgin I
         "St Barthelemy (France)", "St Martin (France)", "Curaçao (Netherlands)", "Bonaire 
         "Saba and Sint Eustatius (Netherlands)", "Sint Maarten (Netherlands)", "Honduras (
         "Colombia (Caribbean)", "Mexico (Atlantic)", "Turks & Caicos Isl. (UK)", "Guyana",
         "Antigua & Barbuda", "Belize", "British Virgin Isl. (UK)", "French Guiana", "Cuba"
         "Anguilla (UK)", "Suriname", "Costa Rica (Caribbean)", "Guatemala (Caribbean)",
         "Panama (Caribbean)", "Bermuda (UK)", "CuraÃ§ao (Netherlands)")

# Brazil
Bra <- c("Brazil (mainland)", "Fernando de Noronha (Brazil)",
         "St Paul and St. Peter Archipelago (Brazil)")

# Canada
Can <-  c("Canada (East Coast)", "Saint Pierre & Miquelon (France)")

# United States
USA <-  c("USA (East Coast)", "USA (Gulf of Mexico)")
```

```
# Northeast Atlantic EEZs
NEA <- c("Canary Isl. (Spain)", "Spain (mainland; Med and Gulf of Cadiz)", "Spain (Northwes
         "Spain (mainland, Med and Gulf of Cadiz)",
         "France (Atlantic Coast)", "Portugal (mainland)", "Azores Isl. (Portugal)")

# Eastern Central Atlantic EEZs
ECA <- c("Madeira Isl. (Portugal)", "Cape Verde", "Equatorial Guinea", "Benin",
          "Sao Tome & Principe ", "Sao Tome & Principe", "Morocco (Central)", "Ghana",
          "Côte d'Ivoire", "CÃ´te d'Ivoire", "Guinea", "Liberia", "Sierra Leone", "Nigeria"
          "Guinea-Bissau", "Senegal", "Cameroon", "Gambia", "Mauritania", "Morocco (South)"
          "Gabon", "Congo; R. of", "Congo, R. of", "Congo (ex-Zaire)")

# Southeastern Atlantic EEZs
SEA <- c("Angola", "Namibia", "South Africa (Atlantic and Cape)",
           "Ascension Isl. (UK)", "Saint Helena (UK)", "Tristan da Cunha Isl. (UK)")

# Southwest Atlantic EEZs
SWA <- c("Trindade & Martim Vaz Isl. (Brazil)", "Uruguay")

# Mediterranean EEZs
Med <- c("Italy (mainland)", "Libya", "Malta", "Syria", "Sicily (Italy)","Sardinia (Italy)"
         "Balearic Islands (Spain)", "Cyprus (North)", "Cyprus (South)",
         "Gaza Strip", "Greece (without Crete)", "Crete (Greece)", "Lebanon",
         "Turkey (Mediterranean Sea)", "Egypt (Mediterranean)", "Israel (Mediterranean)", "
         "Algeria", "Albania", "Croatia", "Montenegro", "Morocco (Mediterranean)",
         "France (Mediterranean)", "Corsica (France)")

# add variable to specify region
d$reg <- "other oceans"
d$reg[which(d$area_name %in% WCA)] <- "Western Central Atlantic EEZs"
d$reg[which(d$area_name %in% Bra)] <- "Brazil EEZ"
d$reg[which(d$area_name %in% Can)] <- "Canadian EEZ"
d$reg[which(d$area_name %in% USA)] <- "United States EEZ"
d$reg[which(d$area_name %in% NEA)] <- "Northeast Atlantic EEZs"
d$reg[which(d$area_name %in% ECA)] <- "Eastern Central Atlantic EEZs"
d$reg[which(d$area_name %in% SEA)] <- "Southeast Atlantic EEZs"
d$reg[which(d$area_name %in% SWA)] <- "Southwest Atlantic EEZs"
d$reg[which(d$area_name %in% Med)] <- "Mediterranean Sea EEZs"
```

Now we will check that we have classified the regions correctly.

```
# check region designation
othlis <- unique(d$area_name[which(d$reg == "other oceans")])
print("EEZs of all other oceans: ")
```

```
[1] "EEZs of all other oceans: "
```

```
othlis
```

```
 [1] "Australia"                        "China"
 [3] "Taiwan"                           "Mayotte (France)"
 [5] "French Polynesia"                 "Hong Kong (China)"
 [7] "Andaman & Nicobar Isl. (India)"   "Indonesia (Eastern)"
 [9] "Japan (main islands)"             "Japan (Daito Islands)"
[11] "Korea (South)"                    "Hawaii Northwest Islands (USA)"
[13] "Nauru"                            "Niue (New Zealand)"
[15] "Northern Marianas (USA)"          "Micronesia (Federated States of)"
[17] "Marshall Isl."                    "Palau"
[19] "Pakistan"                         "Papua New Guinea"
[21] "Philippines"                      "Seychelles"
[23] "Viet Nam"                         "Tokelau (New Zealand)"
[25] "Tonga"                            "Hawaii Main Islands (USA)"
[27] "USA (West Coast)"                 "Wake Isl. (USA)"
[29] "Wallis & Futuna Isl. (France)"    "Samoa"
[31] "Guatemala (Pacific)"              "Kiribati (Gilbert Islands)"
[33] "Mexico (Pacific)"                 "Japan (Ogasawara Islands)"
[35] "Korea (North, Yellow Sea)"        "Korea (North, Sea of Japan)"
[37] "Mauritius"                        "Solomon Isl."
[39] "Christmas Isl. (Australia)"       "Johnston Atoll (USA)"
[41] "Timor Leste"                      "Palmyra Atoll & Kingman Reef (USA)"
[43] "Jarvis Isl. (USA)"                "Howland & Baker Isl. (USA)"
[45] "Indonesia (Central)"              "Indonesia (Indian Ocean)"
[47] "Kiribati (Line Islands)"          "Kiribati (Phoenix Islands)"
[49] "New Caledonia (France)"           "Vanuatu"
[51] "Lord Howe Isl. (Australia)"       "Myanmar"
[53] "Ecuador (mainland)"               "El Salvador"
[55] "Fiji"                             "Malaysia (Peninsula West)"
[57] "Norfolk Isl. (Australia)"         "Peru"
[59] "Pitcairn (UK)"                    "Colombia (Pacific)"
[61] "Costa Rica (Pacific)"             "Nicaragua (Pacific)"
```

```
 [63] "Panama (Pacific)"                    "Thailand (Andaman Sea)"
 [65] "Thailand (Gulf of Thailand)"         "Cook Islands"
 [67] "Kermadec Isl. (New Zealand)"         "Tuvalu"
 [69] "Honduras (Pacific)"                  "American Samoa"
 [71] "Galapagos Isl. (Ecuador)"            "New Zealand"
 [73] "Brunei Darussalam"                   "Malaysia (Sabah)"
 [75] "Malaysia (Sarawak)"                  "Guam (USA)"
 [77] "Clipperton Isl. (France)"            "South Africa (Indian Ocean Coast)"
 [79] "Singapore"                           "Easter Isl. (Chile)"
 [81] "Madagascar"                          "Réunion (France)"
 [83] "Chagos Archipelago (UK)"             "Comoros Isl."
 [85] "Mozambique Channel Isl. (France)"    "Kenya"
 [87] "Mozambique"                          "Somalia"
 [89] "Tanzania"                            "Glorieuse Islands (France)"
 [91] "Yemen (Socotra)"                     "Oman"
 [93] "St Paul & Amsterdam Isl. (France)"   "Iran (Sea of Oman)"
 [95] "Desventuradas Isl. (Chile)"          "India (mainland)"
 [97] "Maldives"                            "Tromelin Isl. (France)"
 [99] "Cambodia"                            "Sri Lanka"
[101] "Malaysia (Peninsula East)"           "Bangladesh"
```

```
print("U.S. EEZs outside the Atlantic Ocean: ")
```

```
[1] "U.S. EEZs outside the Atlantic Ocean: "
```

```
othlis[grep("USA", othlis)]
```

```
 [1] "Hawaii Northwest Islands (USA)"     "Northern Marianas (USA)"
 [3] "Hawaii Main Islands (USA)"          "USA (West Coast)"
 [5] "Wake Isl. (USA)"                    "Johnston Atoll (USA)"
 [7] "Palmyra Atoll & Kingman Reef (USA)" "Jarvis Isl. (USA)"
 [9] "Howland & Baker Isl. (USA)"         "Guam (USA)"
```

```
print("French EEZs outside the Atlantic Ocean: ")
```

```
[1] "French EEZs outside the Atlantic Ocean: "
```

```r
othlis[grep("France", othlis)]
```

```
[1] "Mayotte (France)"                  "Wallis & Futuna Isl. (France)"
[3] "New Caledonia (France)"            "Clipperton Isl. (France)"
[5] "Réunion (France)"                  "Mozambique Channel Isl. (France)"
[7] "Glorieuse Islands (France)"        "St Paul & Amsterdam Isl. (France)"
[9] "Tromelin Isl. (France)"
```

```r
unique(d$area_name[which(d$reg == "Western Central Atlantic EEZs")])   # Western Central EE
```

```
 [1] "Bahamas"
 [2] "Barbados"
 [3] "Bermuda (UK)"
 [4] "Cayman Isl. (UK)"
 [5] "Dominica"
 [6] "Dominican Republic"
 [7] "Grenada"
 [8] "Guadeloupe  (France)"
 [9] "Haiti"
[10] "Jamaica"
[11] "Martinique (France)"
[12] "Montserrat (UK)"
[13] "Puerto Rico (USA)"
[14] "Saint Kitts & Nevis"
[15] "Saint Lucia"
[16] "Saint Vincent & the Grenadines"
[17] "Trinidad & Tobago"
[18] "US Virgin Isl."
[19] "St Barthelemy (France)"
[20] "St Martin (France)"
[21] "Curaçao (Netherlands)"
[22] "Bonaire (Netherlands)"
[23] "Saba and Sint Eustatius (Netherlands)"
[24] "Sint Maarten (Netherlands)"
[25] "Honduras (Caribbean)"
[26] "Colombia (Caribbean)"
[27] "Mexico (Atlantic)"
[28] "Turks & Caicos Isl. (UK)"
[29] "Costa Rica (Caribbean)"
```

```
[30] "Panama (Caribbean)"
[31] "Antigua & Barbuda"
[32] "Belize"
[33] "British Virgin Isl. (UK)"
[34] "Cuba"
[35] "French Guiana"
[36] "Guyana"
[37] "Anguilla (UK)"
[38] "Suriname"
[39] "Venezuela"
[40] "Guatemala (Caribbean)"
[41] "Aruba (Netherlands)"
[42] "Nicaragua (Caribbean)"
```

```
unique(d$area_name[which(d$reg == "Brazil EEZ")])   # Brazil EEZs:
```

```
[1] "Brazil (mainland)"
[2] "Fernando de Noronha (Brazil)"
[3] "St Paul and St. Peter Archipelago (Brazil)"
```

```
unique(d$area_name[which(d$reg == "Canadian EEZ")])   # Canadian EEZs:
```

```
[1] "Canada (East Coast)"              "Saint Pierre & Miquelon (France)"
```

```
unique(d$area_name[which(d$reg == "United States EEZ")])  # United States EEZs:
```

```
[1] "USA (East Coast)"     "USA (Gulf of Mexico)"
```

```
unique(d$area_name[which(d$reg == "Northeast Atlantic EEZs")])   # Northeast Atlantic EEZs:
```

```
[1] "Azores Isl. (Portugal)"
[2] "Spain (mainland, Med and Gulf of Cadiz)"
[3] "Canary Isl. (Spain)"
[4] "Portugal (mainland)"
[5] "France (Atlantic Coast)"
[6] "Spain (Northwest)"
```

```r
unique(d$area_name[which(d$reg == "Eastern Central Atlantic EEZs")])   # Eastern Central At
```

```
 [1] "Cape Verde"                 "Equatorial Guinea"
 [3] "Sao Tome & Principe"        "Madeira Isl. (Portugal)"
 [5] "Togo"                       "Ghana"
 [7] "Guinea"                     "Liberia"
 [9] "Guinea-Bissau"              "Senegal"
[11] "Sierra Leone"               "Morocco (Central)"
[13] "Morocco (South)"            "Côte d'Ivoire"
[15] "Gabon"                      "Benin"
[17] "Gambia"                     "Mauritania"
[19] "Nigeria"                    "Cameroon"
[21] "Congo, R. of"               "Congo (ex-Zaire)"
```

```r
unique(d$area_name[which(d$reg == "Southeast Atlantic EEZs")])   # Southeast Atlantic EEZs:
```

```
[1] "Ascension Isl. (UK)"                "South Africa (Atlantic and Cape)"
[3] "Saint Helena (UK)"                  "Angola"
[5] "Namibia"                            "Tristan da Cunha Isl. (UK)"
```

```r
unique(d$area_name[which(d$reg == "Southwest Atlantic EEZs")])   # Southwest Atlantic EEZs:
```

```
[1] "Trindade & Martim Vaz Isl. (Brazil)" "Uruguay"
```

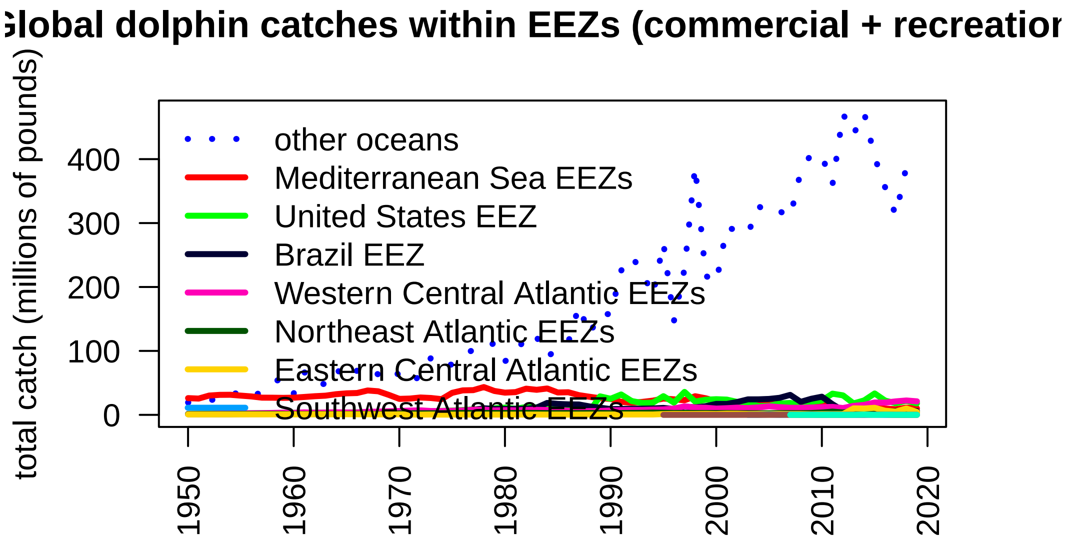
```r
unique(d$area_name[which(d$reg == "Mediterranean Sea EEZs")])   # Mediterranean Sea EEZs:
```

```
 [1] "Italy (mainland)"          "Libya"
 [3] "Malta"                     "Syria"
 [5] "Sicily (Italy)"            "Sardinia (Italy)"
 [7] "Balearic Islands (Spain)" "Tunisia"
 [9] "Algeria"                   "Morocco (Mediterranean)"
[11] "France (Mediterranean)"    "Corsica (France)"
[13] "Croatia"                   "Lebanon"
[15] "Albania"                   "Greece (without Crete)"
[17] "Crete (Greece)"            "Cyprus (South)"
```

3.6 Summarize catch by regional EEZs

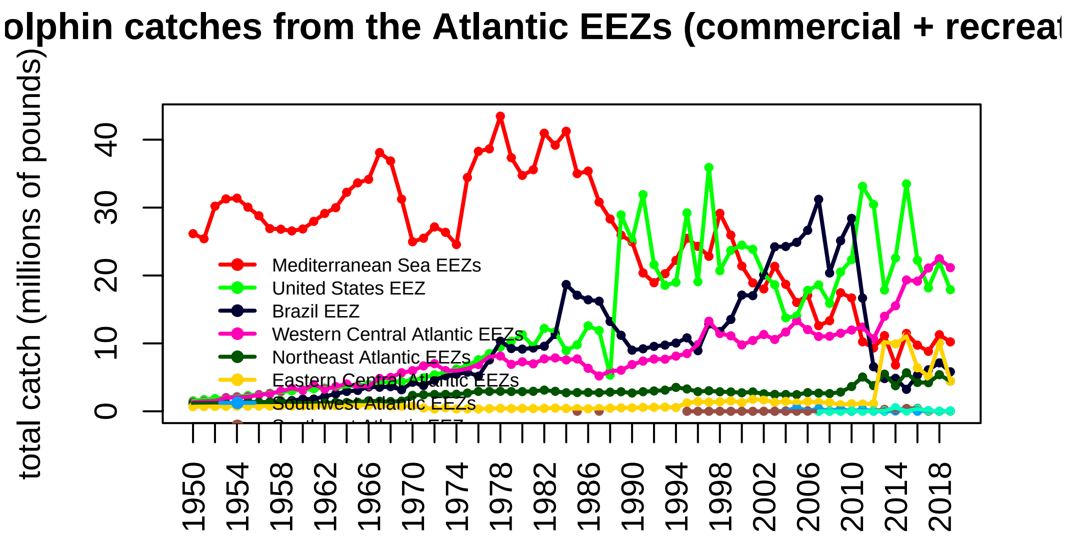
Now we will summarize the catch by region by grouping the catch across the EEZs by region and plot the results. The U.S. landings should be very close to what is reported in Section 2 because the majority of U.S. catch is reported through the various data collection systems that are compiled within Sea Around Us.

```
# summarize EEZ catch by region
tab <- tapply(d$lbs, list(d$year, d$reg), sum, na.rm = T)
#head(tab)
tab2 <- t(tab)
tab3 <- tab2[order(rowSums(tab2, na.rm = T), decreasing = T), ]
tab_glob <- t(tab3)
#head(tab_glob)
yrs <- as.numeric(rownames(tab))

par(mar = c(5, 4, 4, 3), mgp = c(3, 1, 0))
matplot(yrs, tab_glob/10^6, las = 2, type = "l", lty = c(3, rep(1, 9)), lwd = 3, col = glas
        xlab = "", ylab = "total catch (millions of pounds)",
        main = "Global dolphin catches within EEZs (commercial + recreational)")
legend("topleft", colnames(tab_glob), col = glasbey(ncol(tab_glob)),
       lty = c(3, rep(1, 9)), lwd = 3, bty = "n")
```



We can see that the EEZ catches globally are dominated by areas outside of the Atlantic ("other oceans") so we will replot the data after moving this category. Here we can see that most of the catches historically were coming from the Mediterranean Sea, though these catches have been declining drastially since the 1970s. Catches in the United States have showed an opposite pattern, with low historical catches that increased in the 1980s (largely due to the introduction of recreational monitoring). Catches in the Western Central EEZs have increased gradually over time.

```
# remove other oceans
#head(tab_glob)
tab <- tab_glob[, -which(colnames(tab_glob) == "other oceans")]
#head(tab)

matplot(yrs, tab/10^6, las = 2, type = "b", pch = 19, cex = 0.5, lty = 1, col = glasbey(nco
        xlab = "", ylab = "total catch (millions of pounds)", axes = F,
        main = "Dolphin catches from the Atlantic EEZs (commercial + recreational)")
matplot(yrs, tab/10^6, las = 2, type = "l", pch = 19, lty = 1, lwd = 2, add = T, col = glas
axis(1, at = yrs[seq(1, 90, 2)], las = 2); axis(2); box()
legend(1952, 25, colnames(tab), col = glasbey(ncol(tab)+1)[-1], pch = 19, lty = 1, lwd = 2,
```



3.7 Merge high seas with catch with exclusive economic zones

Now we will merge in the high seas catch from the steps above into the EEZ catches to get a better picture of total catch in FAO zones 31 and 21. Recall that the U.S. Gulf and Atlantic catches are not necessarily parsed apart correctly in the public databases which get compiled into Sea Around Us, so these will have the same deficiencies as noted in Section 2. Later we will merge in the corrected data.

```
hs <- read.csv("data/SAU/high_seas_by_year.csv")
names(hs)[1] <- "year"

# check that columns match
#head(tab)
#head(hs)
table(rownames(tab) == hs$year)
```

```
TRUE
  70
```

```r
tab[which(is.na(tab))] <- 0
tab_atl <- cbind(tab, hs[2:3])

# check the merge
head(tab_atl)
```

```
     Mediterranean Sea EEZs United States EEZ Brazil EEZ
1950               26162262           1548245    1127242
1951               25410830           1721403    1060734
1952               30206350           1871779    1128992
1953               31275619           2024535    1111008
1954               31383840           2177865    1090492
1955               30056210           2309815    1190424
     Western Central Atlantic EEZs Northeast Atlantic EEZs
1950                       1345027                 1173441
1951                       1467925                 1187386
1952                       1510406                 1193045
1953                       2001465                 1141504
1954                       2069754                 1269523
1955                       2130341                 1405543
     Eastern Central Atlantic EEZs Southwest Atlantic EEZs
1950                      677552.5                       0
1951                      693004.4                       0
1952                      708466.7                       0
1953                      723939.4                       0
1954                      739422.5                       0
1955                      754915.9                       0
     Southeast Atlantic EEZs Canadian EEZ Atlantic..Northwest
1950                       0            0                   0
1951                       0            0                   0
1952                       0            0                   0
1953                       0            0                   0
1954                       0            0                   0
1955                       0            0                   0
     Atlantic..Western.Central
1950                         0
1951                         0
1952                         0
1953                         0
1954                         0
```

```
1955                         0
```

```r
# separate all WCA catches
unique(d$reg)  # use Western Central, U.S. and Canada
```

```
 [1] "other oceans"                   "Western Central Atlantic EEZs"
 [3] "Brazil EEZ"                     "Eastern Central Atlantic EEZs"
 [5] "Mediterranean Sea EEZs"         "Northeast Atlantic EEZs"
 [7] "United States EEZ"              "Southeast Atlantic EEZs"
 [9] "Southwest Atlantic EEZs"        "Canadian EEZ"
```

```r
dwc <- d[which(d$reg == "Western Central Atlantic EEZs" | d$reg == "United States EEZ" | d$

#unique(dwc$area_name[which(dwc$reg == "United States EEZ")])
#unique(dwc$area_name[which(dwc$reg == "Canadian EEZ")])
#unique(dwc$area_name[which(dwc$reg == "Western Central Atlantic EEZs")])

# relabel U.S. by area
dwc$reg[which(dwc$area_name == "USA (East Coast)")] <- "United States East Coast"
dwc$reg[which(dwc$area_name == "USA (Gulf of Mexico)")] <- "United States Gulf"
dwc$reg[which(dwc$area_name == "Puerto Rico (USA)")] <- "United States Caribbean"
dwc$reg[which(dwc$area_name == "US Virgin Isl.")] <- "United States Caribbean"
#table(dwc$area_name, dwc$reg)

tab_wc <- tapply(dwc$lbs, list(dwc$year, dwc$reg), sum, na.rm = T)
#head(tab_wc)

# reorder S to N
tab_wc <- tab_wc[, c(5, 2, 4, 3, 1)]
head(tab_wc)
```

```
     Western Central Atlantic EEZs United States Caribbean United States Gulf
1950                       1202464                142562.8           309225.3
1951                       1299559                168365.6           339921.3
1952                       1341943                168463.5           370512.4
1953                       1839330                162135.2           400578.5
1954                       1901095                168659.3           433479.6
1955                       1954511                175830.1           463752.0
     United States East Coast Canadian EEZ
```

```
1950                   1239020              NA
1951                   1381482              NA
1952                   1501267              NA
1953                   1623956              NA
1954                   1744386              NA
1955                   1846063              NA
```

```
# merge in the high seas catch - check years match
#head(hs)
table(rownames(tab_wc) == hs$year)
```

```
TRUE
  70
```

```
high_seas <- hs$Atlantic..Western.Central + hs$Atlantic..Northwest
high_seas[high_seas == 0] <- NA
tab_wc <- cbind(tab_wc, high_seas)
head(tab_wc)
```

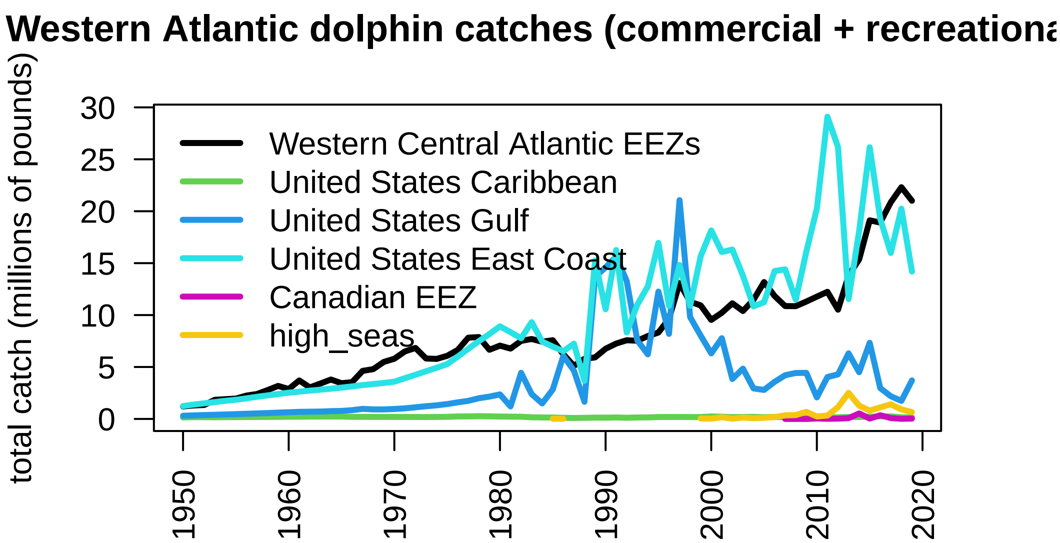
```
     Western Central Atlantic EEZs United States Caribbean United States Gulf
1950                       1202464                142562.8           309225.3
1951                       1299559                168365.6           339921.3
1952                       1341943                168463.5           370512.4
1953                       1839330                162135.2           400578.5
1954                       1901095                168659.3           433479.6
1955                       1954511                175830.1           463752.0
     United States East Coast Canadian EEZ high_seas
1950                  1239020           NA        NA
1951                  1381482           NA        NA
1952                  1501267           NA        NA
1953                  1623956           NA        NA
1954                  1744386           NA        NA
1955                  1846063           NA        NA
```

3.8 Plot Western Atlantic catches

Now we plot the total Western Atlantic catches – both within the EEZs and the high seas – separating by U.S., Canada, and the Western Central Atlantic (i.e., the Wider

Caribbean). Most of the catch is coming from the United States and the 40 combined Caribbean jurisdictions which comprise the Western Central Atlantic.

```
par(mar = c(5, 4, 4, 3), mgp = c(3, 1, 0))
matplot(yrs, tab_wc/10^6, las = 2, type = "l", lty = 1, lwd = 3, col = c(1, 3:7),
        xlab = "", ylab = "total catch (millions of pounds)",
        main = "Western Atlantic dolphin catches (commercial + recreational)")
legend("topleft", colnames(tab_wc), col = c(1, 3:7), lty = 1, lwd = 3, bty = "n")
```



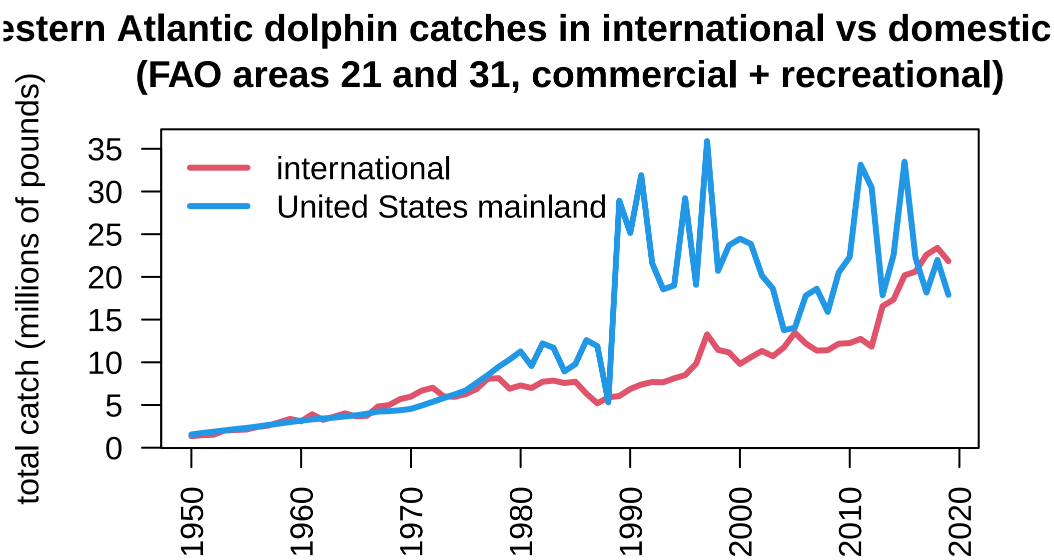
Separating out the mainland United States catches, here are the international and domestic (mainland) catches plotted over time.

```
tab_int <- tab_wc[, c(1, 2, 5, 6)]
head(tab_int)
```

```
     Western Central Atlantic EEZs United States Caribbean Canadian EEZ
1950                       1202464                142562.8           NA
1951                       1299559                168365.6           NA
1952                       1341943                168463.5           NA
1953                       1839330                162135.2           NA
1954                       1901095                168659.3           NA
```

```
1955                        1954511                 175830.1          NA
     high_seas
1950        NA
1951        NA
1952        NA
1953        NA
1954        NA
1955        NA
```

```
int_catch <- round(rowSums(tab_int, na.rm = T), 2)
dom_catch <- rowSums(tab_wc[, c(3, 4)], na.rm = T)

matplot(yrs, cbind(int_catch, dom_catch)/10^6, las = 2, type = "l", lty = 1, lwd = 3, col =
        xlab = "", ylab = "total catch (millions of pounds)",
        main = "Western Atlantic dolphin catches in international vs domestic waters\n(FAO
legend("topleft", c("international", "United States mainland"), col = c(2, 4), lty = 1, lwd
```

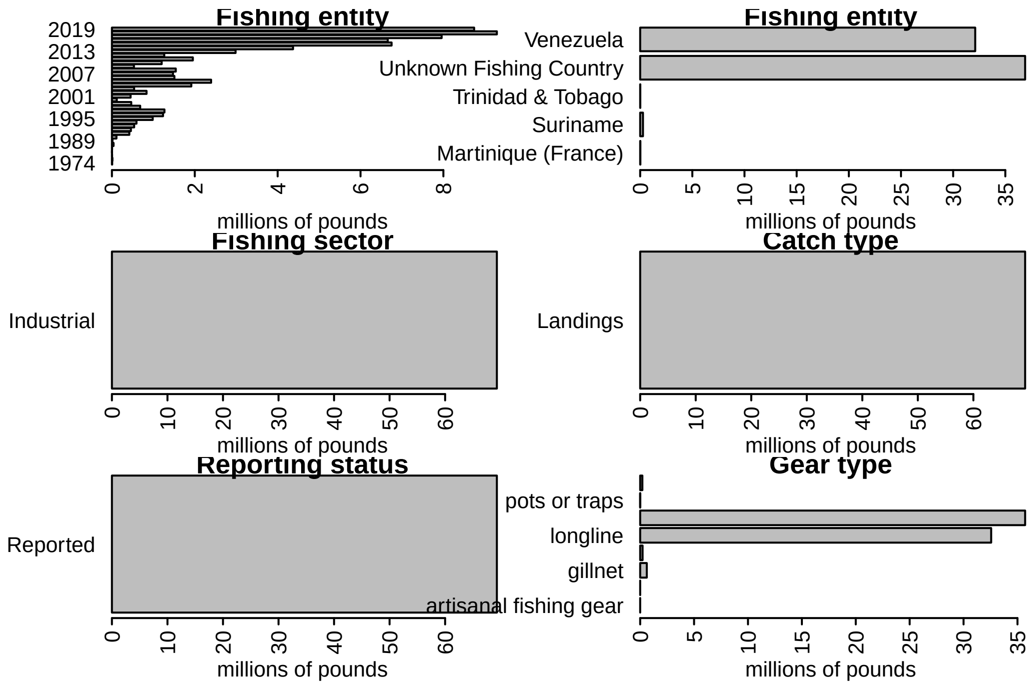



```
# output international catches
write.csv(data.frame(int_catch), file = "data/international_catches.csv", row.names = T)
```

3.9 Look at Venezuela catches in detail

Note in the plot above a rapid increase in international catches within the Western Atlantic. This is driven by a sudden increase in catches reported within Sea Around Us within the EEZ. We investigate these catches in detail as it seems odd that a relatively small fleet would be able to ramp up effort so rapidly. Data can be downloaded on a country-specific basis, the catches by taxon in the waters of Venezuela are accessed here.

```
rm(list = ls())  # clear the workspace

# download the data for waters of Venezuela
ven <- read.csv("data/SAU/SAU EEZ 862 v50-1.csv")

# subset dolphinfish from the data
v <- ven[which(ven$common_name == "Common dolphinfish"), ]

#apply(v[1:13], 2, table)

#convert to pounds
v$pounds <- v$tonnes * 1000 * 2.20462

# look at the characteristics of the data
par(mar = c(5, 10, 1, 1), mfrow = c(3, 2), mex = 0.5)
#barplot(tapply(v$pounds/10^6, v$area_name, sum, na.rm = T), las = 2, xlab = "millions of p
#barplot(tapply(v$pounds/10^6, v$area_type, sum, na.rm = T), las = 2, xlab = "millions of p
barplot(tapply(v$pounds/10^6, v$year, sum, na.rm = T), las = 2, xlab = "millions of pounds"
barplot(tapply(v$pounds/10^6, v$fishing_entity, sum, na.rm = T), las = 2, xlab = "millions 
barplot(tapply(v$pounds/10^6, v$fishing_sector, sum, na.rm = T), las = 2, xlab = "millions 
barplot(tapply(v$pounds/10^6, v$catch_type, sum, na.rm = T), las = 2, xlab = "millions of p
barplot(tapply(v$pounds/10^6, v$reporting_status, sum, na.rm = T), las = 2, xlab = "million
barplot(tapply(v$pounds/10^6, v$gear_type, sum, na.rm = T), las = 2, xlab = "millions of po
```



Note that a large portion of the catch in the database is listed as coming from an "Unknown Fishing Country." We extract these catches and further investigate the characteristics.

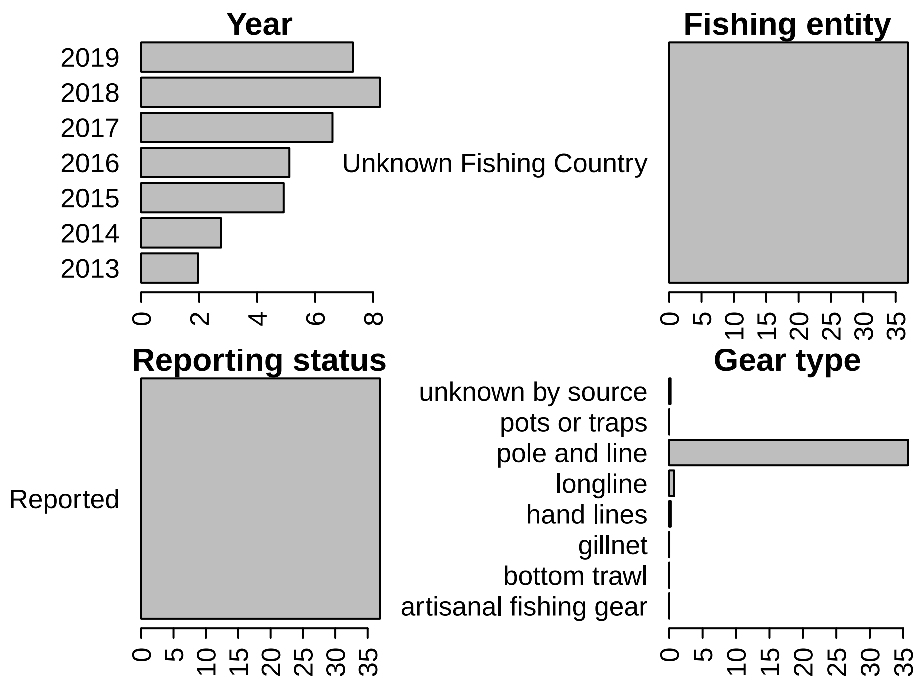
```
# extract the Unknown Fishing Country catch
v1 <- v[which(v$fishing_entity == "Unknown Fishing Country"), ]
apply(v1[1:15], 2, table)
```

```
$area_name

Venezuela
       39

$area_type

eez
 39

$data_layer

Assigned Tuna RFMO catch
```

```
                      39

$uncertainty_score
< table of extent 0 >

$year

2013 2014 2015 2016 2017 2018 2019
   5    5    5    6    7    5    6

$scientific_name

Coryphaena hippurus
                 39

$common_name

Common dolphinfish
                39

$functional_group

Large pelagics (>=90 cm)
                      39

$commercial_group

Perch-likes
         39

$fishing_entity

Unknown Fishing Country
                     39

$fishing_sector

Industrial
        39

$catch_type
```

```
Landings 
      39 

$reporting_status

Reported 
      39 

$gear_type

artisanal fishing gear           bottom trawl                gillnet 
                     1                      5                      7 
            hand lines               longline          pole and line 
                     3                      4                      7 
         pots or traps      unknown by source 
                     5                      7 

$end_use_type

Direct human consumption 
                      39 
```

```
# look at characteristics
par(mar = c(3, 10, 1, 5), mfrow = c(2, 2), mex = 0.6)
barplot(tapply(v1$pounds/10^6, v1$year, sum, na.rm = T), las = 2, xlab = "millions of pound
barplot(tapply(v1$pounds/10^6, v1$fishing_entity, sum, na.rm = T), las = 2, xlab = "million
barplot(tapply(v1$pounds/10^6, v1$reporting_status, sum, na.rm = T), las = 2, xlab = "milli
barplot(tapply(v1$pounds/10^6, v1$gear_type, sum, na.rm = T), las = 2, xlab = "millions of 
```

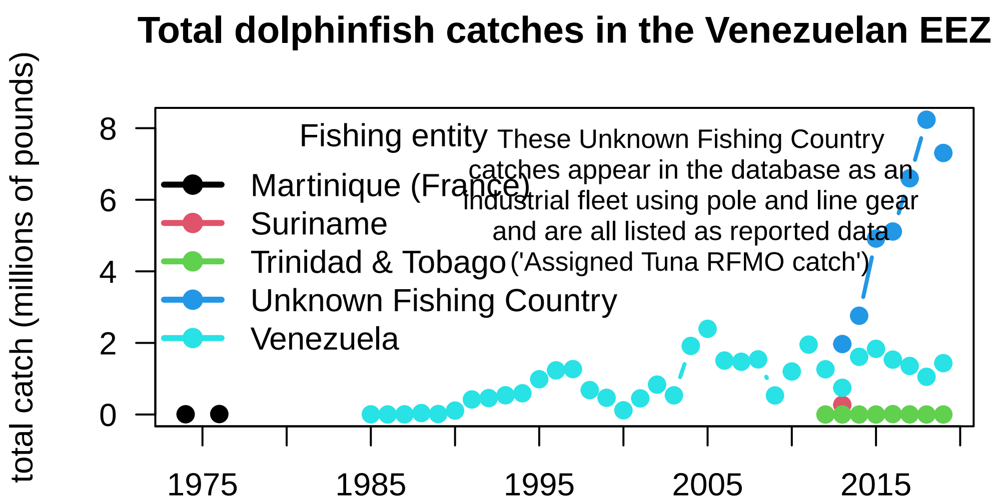


Here is another look at the catches within the venezuelan EEZ, separated by fishing entity.

```
# summarize Venezuela EEZ catches by fishing entity
tab <- tapply(v$pounds, list(v$year, v$fishing_entity), sum, na.rm = T)
head(tab)
```

```
     Martinique (France) Suriname Trinidad & Tobago Unknown Fishing Country
1974            7320.429       NA                NA                      NA
1976           12814.641       NA                NA                      NA
1985                  NA       NA                NA                      NA
1986                  NA       NA                NA                      NA
1987                  NA       NA                NA                      NA
1988                  NA       NA                NA                      NA
     Venezuela
1974        NA
1976        NA
1985  4699.563
1986  2229.484
1987  4585.610
1988 39960.149
```

```
matplot(rownames(tab), tab/10^6, las = 2, type = "b", lty = 1, lwd = 2, pch = 19, col = 1:5
        xlab = "", ylab = "total catch (millions of pounds)", axes = F,
        main = "Total dolphinfish catches in the Venezuelan EEZ")
axis(1, at = seq(1950, 2020, 5)); axis(2, las = 2); box()
legend("topleft", colnames(tab), col = 1:5, pch = 19, lty = 1, lwd = 3, bty = "n", title =

text(2004, 6, cex = 0.8,
     "These Unknown Fishing Country\ncatches appear in the database as an\nindustrial fleet
```



The very substantial increase in international Western Atlantic dolphin catches can be attributed solely to this Unknown Fishing Country catches which appear in the database beginning in 2013. The documentation of the catch reconstruction can be found in the Sea Around Us documentation files (see Mendoza et al. 2015). We spoke to Jeremy Mendoza and other experts who were involved in the reconstruction of domestic catches in Venezuela, and they noted that the "Assigned tuna RFMO catch" actually emerge from spatially reshuffled landing reports to ICCAT, according to the methods of Coulter et al. (2020). In this work, the Sea Around Us team developed a global spatial reallocation of landings of tunas and associated species that were reported to ICCAT. In short, this work uses the fraction of reported spatially disaggregated data to reallocate all data reported to ICCAT. This step is necessary because there are not many countries that report spatial data to ICCAT in the region, and this method may lead to a significant reallocation of landings from the Caribbean to the Venezuelan EEZ, where the local pole and line fishery is significant (J. Mendoza, personal communication). Thus, it is likely that landings from other EEZs, that were included in the reconstructed catch data layer, could have been reassigned to the Venezuelan EEZ.

Because there may be some overlap between the reconstructed data used in the national EEZ estimates, and the ICCAT reporting that was reassigned, there could potentially be a “double counting” of these catch estimates in the Sea Around Us database. These Unknown Fishing Country catches that lead to a large increase in total estimated catches within the Venezuelan EEZ should be treated with caution at this time, until we can verify with Sea Around Us database managers that the spatial disaggregation method used has not led to an overestimate.

3.10 References

Pauly D., Zeller D., Palomares M.L.D. (Editors), 2020. Sea Around Us Concepts, Design and Data (seaaroundus.org).

4 NOAA Fisheries landings data

Here we will explore commercial and recreational landings data accessed through the FOSS (Fisheries One Stop Shop) data portal. The advantage of this portal is that commercial and recreational data are available side-by-side in the same units (pounds landed).

4.1 Data access and upload

Data are downloaded from the NOAA Fisheries FOSS site. For the Data Set, we choose both commercial and recreational. For the Years, we select all years from 1980 to the present. For NMFS Region, we select the "NMFS Regions" button and then select the Gulf, South Atlantic, Middle Atlantic and New England from the box. For the Species, we select "Dolphinfish" and "Dolphinfish**." For Report Format, we choose TOTALS BY YEAR/REGION. Hitting the "RUN REPORT" button then generates the data needed for the subsequent analysis, which can be downloaded as a .csv file under the "Actions" button.

We then read in the data file and clean it for analysis (removing the commas from the numbers so that R recognizes the values as numbers). We output the file header to make sure it was read in and that the numbers were converted correctly.

```
# clear workspace
rm(list = ls())

# read in data file
d <- read.table("data/FOSS_landings.csv", header = T, sep = ",", skip = 1)
head(d)
```

```
  Year     Region.Name    Pounds Dollars   Collection Metric.Tons
1 1980            Gulf    94,984  55,676   Commercial          43
2 1980 Middle Atlantic       400     256   Commercial           0
3 1980     New England       200      54   Commercial           0
```

```
4 1980  South Atlantic    76,042  45,852   Commercial           34
5 1981            Gulf    73,075  46,951   Commercial           33
6 1981            Gulf 1,559,544         Recreational          707
```

```
# take commas out of file --------------------------
d$Pounds <- as.numeric(gsub(",", "", d$Pounds))
d$Dollars <- as.numeric(gsub(",", "", d$Dollars))
d$Metric.Tons <- as.numeric(gsub(",", "", d$Metric.Tons))
head(d)
```

```
  Year     Region.Name  Pounds Dollars   Collection Metric.Tons
1 1980            Gulf   94984   55676   Commercial          43
2 1980 Middle Atlantic     400     256   Commercial           0
3 1980     New England     200      54   Commercial           0
4 1980  South Atlantic   76042   45852   Commercial          34
5 1981            Gulf   73075   46951   Commercial          33
6 1981            Gulf 1559544      NA Recreational         707
```

4.2 Summarize the landings

We then summarize the landings (in pounds) by year, sector, and region. The result is a new data frame with each row representing a year and each column representing the total landings for each region and sector.

```
# compile by year for sectors and regions ---------------
d$Collection <- factor(d$Collection, levels = c("Recreational", "Commercial"))
tab <- tapply(d$Pounds, list(d$Year, d$Collection, d$Region.Name), sum, na.rm = T)
tab[is.na(tab)] <- 0
#head(tab)

dat <- data.frame(cbind(tab[, , "Gulf"], tab[, , "South Atlantic"],
                        tab[, , "Middle Atlantic"], tab[, , "New England"]))
names(dat) <- c("rec_Gulf", "com_Gulf", "rec_SA", "com_SA", "rec_MA", "com_MA", "rec_NE", "
dat
```

```
     rec_Gulf com_Gulf   rec_SA  com_SA  rec_MA com_MA  rec_NE com_NE
1980        0    94984        0   76042       0    400       0    200
1981  1559544    73075  8421898   59037       0      0       0      0
```

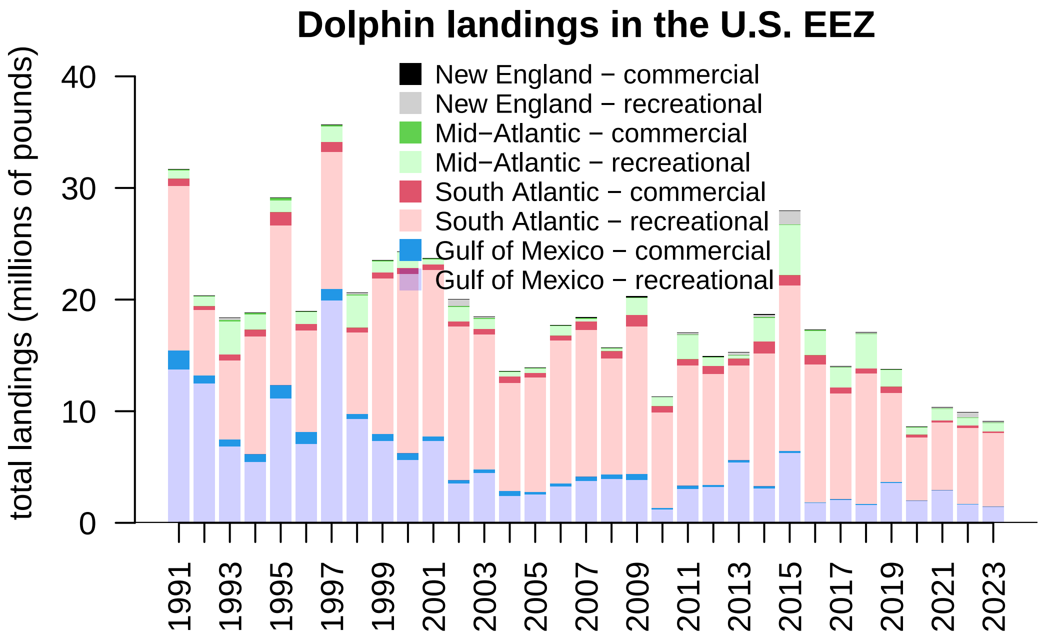
1982	4197503	162457	7573035	143960	2097	200	0	0
1983	2162936	190808	9134022	127986	45927	1400	0	400
1984	1189573	282976	7265430	158901	0	1700	0	400
1985	2542089	269477	6684821	142684	113666	5000	0	4800
1986	5540003	492212	6190187	189747	125653	4200	0	200
1987	4209638	403703	6883546	212905	121433	13400	0	1100
1988	1014312	491610	3134070	212709	130240	26600	2196	17800
1989	12551461	1029628	13273504	426392	1386752	81700	437	15300
1990	13389540	1098819	9283973	514980	624928	69106	2714	14233
1991	13760600	1674124	14756442	650333	722090	90722	6294	9816
1992	12481422	714369	5893700	326831	860112	74423	1411	10315
1993	6873820	596967	7077105	520512	3001335	99420	200927	27069
1994	5448709	679903	10554836	618614	1400635	101044	12586	12328
1995	11142952	1177325	14318412	1164608	1096473	198425	24061	13811
1996	7086006	1047020	9128895	548682	1077364	43581	0	7788
1997	19958088	989648	12284366	877303	1422057	105559	0	13265
1998	9324186	422501	7312090	406368	2920384	86502	120322	11553
1999	7329953	612445	13973364	504263	1005852	98927	4458	5990
2000	5641491	582576	16049049	537077	1433991	36048	0	6545
2001	7355385	368503	14945284	470897	477081	51342	5189	29134
2002	3513226	290601	13794853	436472	1295470	83534	600948	14836
2003	4481937	310836	12112718	459351	929763	46851	129513	21275
2004	2402315	435341	9707678	555973	427145	31916	0	29042
2005	2537132	207695	10299670	376989	437472	19703	0	19982
2006	3290474	224668	12805952	433121	887684	27724	0	19645
2007	3754513	370397	13164377	740255	241212	79832	19542	44707
2008	3937153	384954	10417536	627939	264017	57764	0	16328
2009	3860527	496701	13251102	1007355	1538112	55271	0	92010
2010	1216198	111790	8543542	572031	813738	35869	0	14939
2011	3023914	309324	10774812	543811	2173136	15376	187289	26587
2012	3229090	146256	9983276	704240	755558	44918	26892	46171
2013	5425636	187966	8483569	583556	302421	41936	226988	18564
2014	3074373	201943	11925345	1041856	2129222	58868	183673	61505
2015	6286577	154133	14821772	920454	4501793	36190	1198628	37808
2016	1777348	37411	12357817	828133	2212513	61336	16499	15747
2017	2064106	65141	9457460	532973	1820541	16179	48607	30969
2018	1626095	56188	11713866	435059	3104018	20514	97107	12667
2019	3567170	83956	7967978	563132	1518629	48224	66	16918
2020	1952241	26332	5653294	260311	679556	14040	0	1880
2021	2893817	26570	6061024	178822	1051785	25318	123038	11421
2022	1664771	15906	6845573	182798	708494	17360	466897	9131

```
2023   1427518     13305   6630236   125302   734866   28467   122550    8310
```

4.3 Plot the summary

Now we plot the data, using similar color schemes by region. Because the recreational data have high imprecision prior to the 1990s we plot from 1990 to the last year of data.

```r
# select years ---------------------
dat <- dat[which(rownames(dat) > 1990), ]

# define colors and plot ------------------
cols <- c("#0000FF30", 4, "#FF000030", 2, "#00FF0030", 3, "#00000030", 1)

par(mar = c(4, 4, 1, 0), mgp = c(2.5, 1, 0))
yrs <- as.numeric(rownames(dat))

b <- barplot(t(dat)/10^6, names.arg = yrs, col = cols, ylim = c(0, 43),
             border = NA, las = 2, ylab = "total landings (millions of pounds)",
             main = "Dolphin landings in the U.S. EEZ")
legend("top", c("New England - commercial", "New England - recreational",
                "Mid-Atlantic - commercial", "Mid-Atlantic - recreational",
                 "South Atlantic - commercial", "South Atlantic - recreational",
                 "Gulf of Mexico - commercial", "Gulf of Mexico - recreational"),
    col = cols[8:1], pch = 15, pt.cex = 1.5, bty = "n", y.intersp = 0.9, cex = 0.85)
axis(1, at = b, lab = rep("", length(b)), pos = 0)
abline(h=0)
```



4.4 Updated Gulf/Atlantic parsing of data

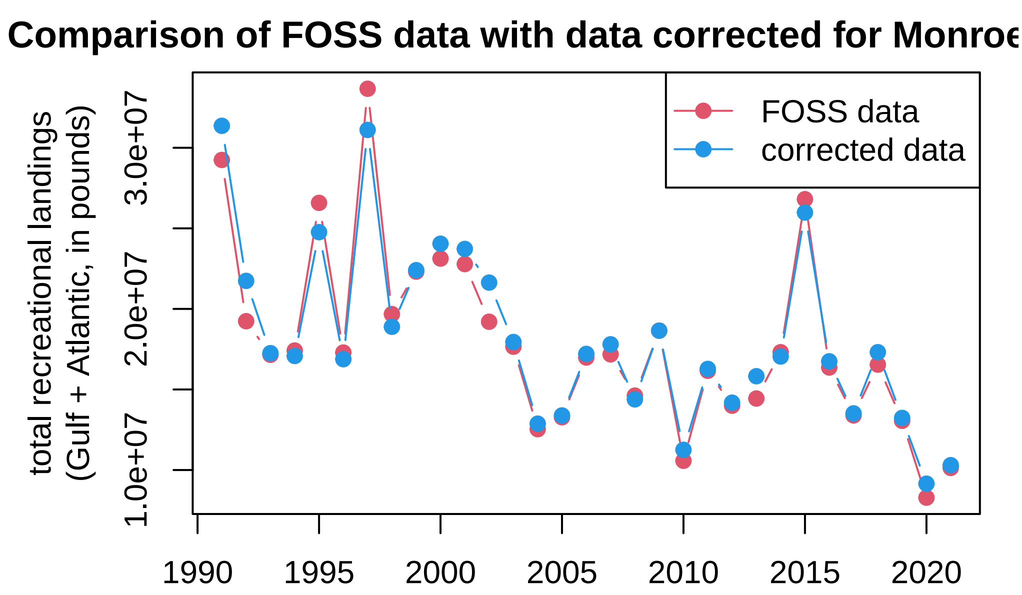
There are some issues with the way that the FOSS system splits out Gulf and Atlantic catches by Monroe County, which is on the border between the two jurisdictions. We received data from Mike Larkin (SERO) which has the correct split out of data for Monroe County between the Gulf and Atlantic. We compare the total recreational catch for the Gulf and Atlantic between the FOSS data and the updated data. The numbers are very close; the average difference from the FOSS data is 2%.

```
ml <- read.table("data/RecreationalDolphinLandings_SAandGOM_MLarkin.csv", header = T, sep =
ml <- ml[which(ml$Year > 1990), ]

# select same years for FOSS data set
dat <- dat[which(yrs <= max(ml$Year)), ]

# check that years match
summary(rownames(dat) == ml$Year)
```

```
   Mode    TRUE
```

```
logical      31
```

```
yrs <- as.numeric(rownames(dat))

# check that totals match
old <- dat$rec_Gulf + dat$rec_SA + dat$rec_MA + dat$rec_NE
new <- ml$GOM + ml$ATL

# plot totals
par(mar = c(4, 6, 2, 1), mgp = c(2.5, 1, 0))
matplot(yrs, cbind(old, new), xlab = "", lty = 1, col = c(2, 4), type = "b", pch = 19,
        ylab = "total recreational landings\n(Gulf + Atlantic, in pounds)",
        main = "Comparison of FOSS data with data corrected for Monroe County")
legend("topright", c("FOSS data", "corrected data"), col = c(2, 4), pch = 19, lty = 1)
```



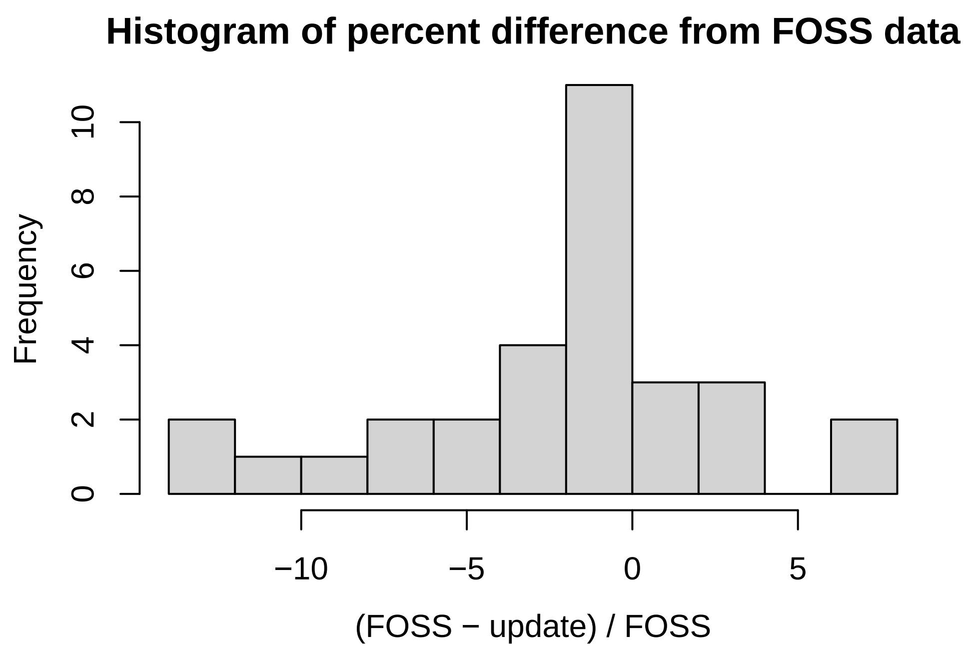

```
# average % difference between FOSS and updated data
mean((old - new)/old*100)
```

```
[1] -2.023886
```

```
# min and max % difference between FOSS and updated data
range((old - new)/old*100)
```

```
[1] -13.008552   7.587789
```

```
hist((old - new)/old*100, main = "Histogram of percent difference from FOSS data", breaks =
     xlab = "(FOSS - update) / FOSS")
```




```
# replace FOSS data with update data
datnew <- dat
datnew$rec_Gulf <- ml$GOM
datnew$rec_SA <- ml$ATL - (datnew$rec_MA + datnew$rec_NE)

# old data set
tail(dat)
```

```
     rec_Gulf com_Gulf   rec_SA com_SA  rec_MA com_MA rec_NE com_NE
2016  1777348    37411 12357817 828133 2212513  61336  16499  15747
2017  2064106    65141  9457460 532973 1820541  16179  48607  30969
```

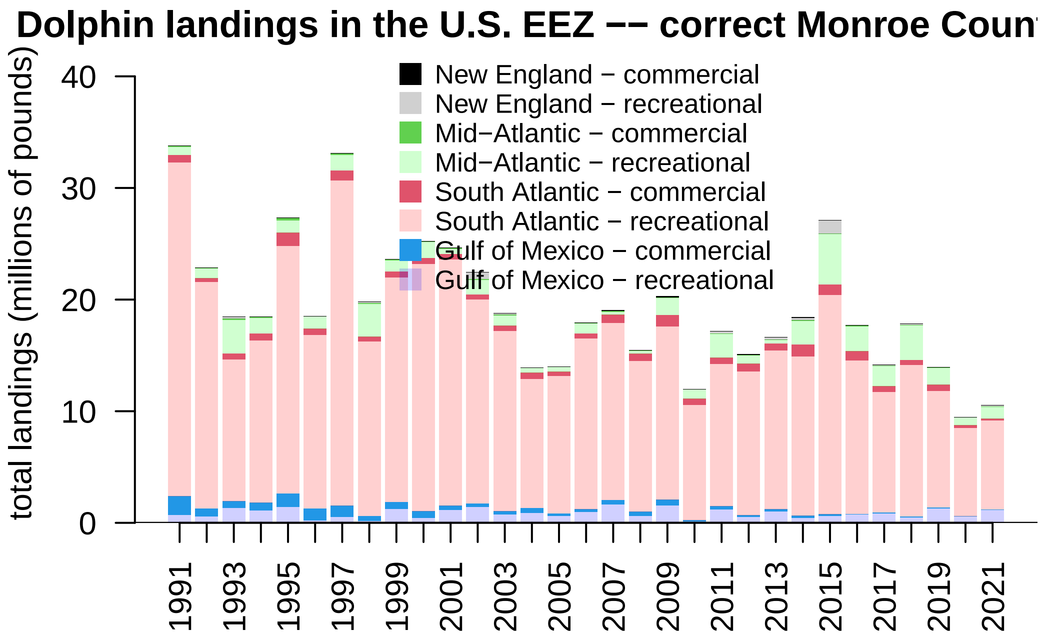
```
2018  1626095    56188 11713866 435059 3104018  20514  97107  12667
2019  3567170    83956  7967978 563132 1518629  48224     66  16918
2020  1952241    26332  5653294 260311  679556  14040      0   1880
2021  2893817    26570  6061024 178822 1051785  25318 123038  11421
```

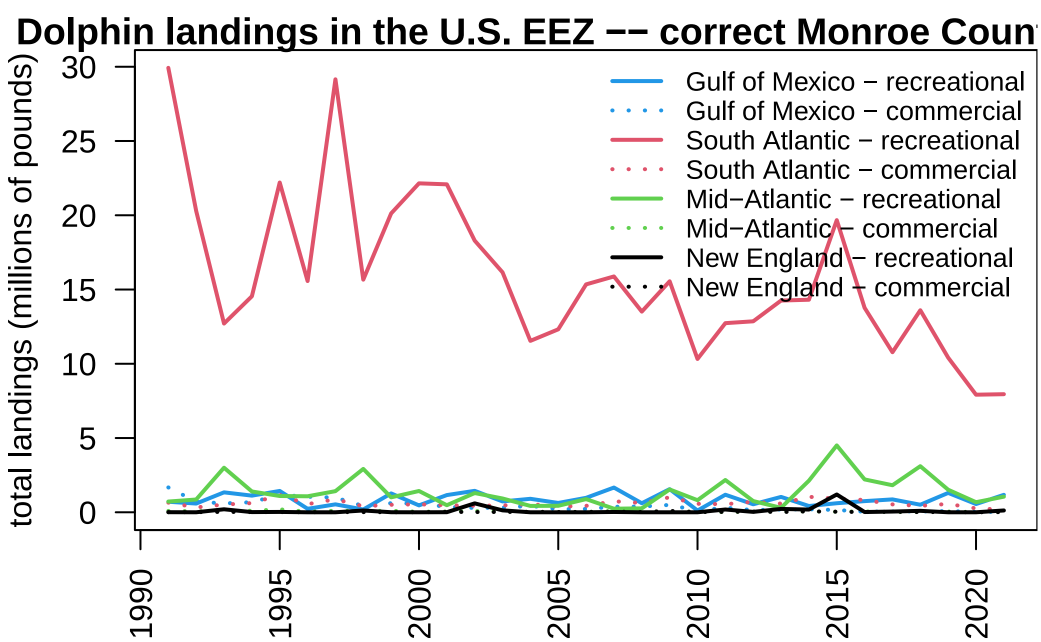
```r
# updated data set
tail(datnew)
```

```
     rec_Gulf com_Gulf   rec_SA com_SA  rec_MA com_MA rec_NE com_NE
2016   751499    37411 13766851 828133 2212513  61336  16499  15747
2017   868489    65141 10777104 532973 1820541  16179  48607  30969
2018   508570    56188 13603875 435059 3104018  20514  97107  12667
2019  1304865    83956 10410599 563132 1518629  48224     66  16918
2020   555695    26332  7916891 260311  679556  14040      0   1880
2021  1174412    26570  7950716 178822 1051785  25318 123038  11421
```

4.5 Plots with updated data

We replace the FOSS values for Gulf and South Atlantic recreational catches with these new values and re-plot the full data set. We can also plot the landings as a line graph so that the relative magnitudes of catch across region and sector are more easily compared.


Dolphin landings in the U.S. EEZ –– correct Monroe Coun
total landings (millions of pounds)
0
10
20
30
40
New England − commercial
New England − recreational
Mid−Atlantic − commercial
Mid−Atlantic − recreational
South Atlantic − commercial
South Atlantic − recreational
Gulf of Mexico − commercial
Gulf of Mexico − recreational
1991
1993
1995
1997
1999
2001
2003
2005
2007
2009
2011
2013
2015
2017
2019
2021


Dolphin landings in the U.S. EEZ –– correct Monroe Coun
total landings (millions of pounds)
0
5
10
15
20
25
30
Gulf of Mexico − recreational
Gulf of Mexico − commercial
South Atlantic − recreational
South Atlantic − commercial
Mid−Atlantic − recreational
Mid−Atlantic − commercial
New England − recreational
New England − commercial
1990
1995
2000
2005
2010
2015
2020

5 SST

6 Analysis of ocean temperature trends

Coming soon